Finite Observer Consensus as a Reconstruction Principle: Normal Forms, the Standard Model Lie Type, and a Route to the Einstein Field Equation

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Abstract

Observer Patch Holography proposes that public physics is reconstructed by finite self-reading patches whose records agree across typed overlaps. Twelve-port carriers federate over spherical support. A signed-record response branch has an exact rank-three quotient, and the celestial S^2 supports Lorentz-frame kinematics. Convergent repair selects protected public normal forms, while authenticated provenance generates finite informational posets with canonical source height. Across their stated branches, the constructions give density-operator quantum probability, the exact Tsirelson bound and no-cloning, the Standard Model gauge Lie type, an anomaly-free fifteen-state generation, and the conditional Einstein equation up to a cosmological constant.

A supplied charged-scalar/Maxwell action provides a calculable matter model. For positive mass squared, nonnegative quartic coupling, smooth Neumann references and Ritz-projected initial data, its real zero-current sector has first-order spatial trajectory convergence to nonlinear quartic-wave solutions on uniform conforming refinements. On the fixed volume mesh, the full reduced interacting kinetic metric is complete. Its declared Laplace–Beltrami Hamiltonian has a unique self-adjoint closure, a neutral Hilbert sector and explicit normalized states.

Conservative source-record populations with a declared local read law converge to flat $(3+1)$ causal order and volume; interval counts recover proper-time ratios. A declared scalar action on the same source addresses has classical and Fock-space continuum comparisons, with certified compact-detector errors smaller than the induced signal throughout a nonzero time window.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its intrinsic computation through local events, records, and responses to admissible interventions. The correspondence must preserve those dependencies; no external host or additional time in which the history is played is required. A candidate must specify the history domain and closure operator; the schematic equation does not construct or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence,

geometric domain, action, representation, or clock remains a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Time conventions. Host runtime, solver counters, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks remain declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

Keywords: foundations of physics; Observer Patch Holography; finite reconstruction; causal sets; quantum records; Einstein field equation; gauge structure.

1 Introduction and claim boundary

Quantum probability, Lorentzian kinematics, gravitational dynamics, and gauge structure enter standard descriptions through distinct mathematical inputs [20, 21, 22, 63, 72]. Observer Patch Holography asks whether they can instead arise as different public readouts of one finite observer architecture.

The primitive object is an observer patch: a finite subsystem with local state, an observable boundary, readback, durable records, and authorized repair moves. At each regulator, local carriers realize observer patches and federate through typed overlaps. Each carrier has twelve primitive boundary ports with oriented icosahedral incidence. The screen is a federation-level construction: its overlap nerve maps with degree one to a refinement tower with oriented S^2 support. Local carrier boundary, federation nerve, and spherical support are distinct parts of the construction.

OPH identifies an objective fact with a record that survives every authorized comparison. A patch begins with a private, repairable record. Patches compare their readbacks on shared boundaries and accept only repairs that preserve protected data. When the named repair relation terminates, satisfies the local diamond, and is complete, every maximal schedule from a fixed initial quotient state reaches the same consistent normal form. The resulting protected record is idempotent, schedule independent, and insensitive to presentation data hidden by the quotient. OPH interprets this public normal form as emergent objective reality. Authenticated read-after-write provenance generates a finite informational poset with an exact longest-parent-chain height, and the rank-three positive source quotient defines the four-dimensional ambient carrier $\mathbb{R} \oplus V_{\mathrm{src}}$, with a Lorentz form and S^2 null-direction space, into which every finite log admits an injective forward-causal placement. That placement is neither source-selected nor order-reflecting; physical causal identification, the direct causal-set route, and smooth Einstein promotion each require the separately premised refinement family of Section 6.

Observer freedom is constrained before a state is selected. A1 fixes the accessible hardware, including local port incidence, overlap interfaces, the complete reversible-response tangent, and the spherical support; A2 fixes agreement on shared operational meaning and requires proper carrier rechartings to be implemented from inside that response; A3 selects the least informative state within the feasible set left by those constraints. On the signed-record response branch the normalized repair response has an intrinsic rank-three Gram quotient, so the public carrier-position

readout leaves observers exactly three independent directions to recover, authorized rechartings preserve their rank, and a faithful physical-position realization must preserve this dimension.

Within the closure hypothesis, the twelve-port carrier and S^2 support are fixed facts of one self-consistent universe. Internal observers encounter them through completed records, can infer the hardware and protocol from their record history, and can later construct a realization that returns the same invariant readouts; the complete structure has no beginning at which the architecture is chosen. The equation $T(\mathfrak{U}) = \mathfrak{U}$ states this universe-level closure hypothesis. Observer record order compares completed records while well-founded repair descent proves termination. No preferred global clock is required, and reconstruction followed by construction in observer record order sends no information toward earlier records. Section 1.5 develops the hypothesis.

The physical thesis is that, in one inhabited common realization, these protected public readouts are the observables of our universe. The supporting results have different logical types. Finite theorems recover density-operator probability, Lüders conditioning, the Tsirelson ceiling, no-cloning, the intrinsic rank-three position carrier, the local Standard Model gauge Lie type, and an anomaly-free fifteen-state generation; reconstruction contracts give Lorentz-frame kinematics and the conditional Einstein equation when their stated geometric and physical antecedents hold; closure diagnostics land close to the measured fine-structure and cosmological constant coordinates; and a frozen prospective propagation branch fixes its first anisotropic harmonic at angular rank six. The full physical thesis requires one common realized tower carrying all of these readouts, including four-dimensional event geometry.

The case for OPH is the reach of one observer architecture: protected normal forms define shared records, complete reversible response with endogenous transport fixes the local gauge Lie type, and one supplied matter action supports both real-sector continuum trajectories and an interacting Hilbert space, connecting objects that ordinary physical models introduce separately. Each link has an explicit input and an independently inspectable output. The closure hypothesis supplies the timeless self-referential reading; A1 fixes patch hardware, A2 fixes agreement and endogenous transport, and A3 selects the least-informative compatible state; named repair and propagation laws support finite theorems and certified computations, while termination, confluence and a physical kinetic operator require their own premises. Each result also declares its local mathematical hypotheses and empirical inputs; the three postulates are not a complete premise list for every result. Physical realization maps identify records, currents, clocks, stress and scales with laboratory or spacetime objects; and spherical support gives Lorentz symmetry, while realized null balance, Ward conservation, the Bianchi identity and an independent scale give the Einstein implication.

Table 2 separates these implications from diagnostics and prospective tests. Companion papers give the longer proofs and executable evidence [1, 5, 6, 7, 8, 13, 11, 2]. Each step is identified as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Table 1 gives the seven epistemic claim classes used throughout.

Throughout, "source" means computed from the declared finite source specification, including every explicitly named source law, with no measured physical input. "Source-only" marks a quantity whose entire dependency cone is of that kind. A physical realization map is not source-only merely because the finite branch calculation is target-clean.

The distinction is mathematical: a finite implication and a physical realization of its antecedents are different claims. Machine-readable receipts record that distinction and reject a physical verdict when a required map is absent.

Boundary 1.1. This paper claims no completed derivation of particle masses, no source-only de-

Table 1: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

termination of the fine-structure constant, no completed cosmology, and no claim that formal verification establishes physical truth. Quantitative surfaces that consume measured values are labeled diagnostics. The primitive twelve-port propagation branch is a frozen prospective conditional prediction. Section 13 states its decision rule and the physical premises that set the scope of any verdict.

Definition 1.2 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

Throughout this paper, a *contract* is a named, typed theorem interface recording the inputs, domains, maps, normalization conventions, premises, outputs, and realization obligations for one implication; it keeps the full dependency surface visible and introduces no additional physical law.

1.1 Principal contributions

Finite records connect to geometry, matter and dynamics; Table 2 states hypotheses and conclusions. The normal-form layer gives a canonical partial normalizer, sharp stability and refinement bounds, and complexity barriers; a transactional local diamond yields schedule-independent public records, bounded waste controls attempt cost, and certified read-from provenance supplies a finite causal order. The carrier layer returns an intrinsic rank-three position readback and forces the gauge Lie type, with a conditional $\mathbb{Z}_6$ matter image and exhaustive exterior selection. The quantum-record layer derives Born weights from finite effect valuations and carries Lüders conditioning, Tsirelson's bound and the copying obstruction on a declared algebra-state pair. The dynamical layer includes a controlled nonlinear continuum limit of a supplied charged-scalar/Maxwell action, whose real zero-current sector converges at first order in the energy norm on a uniform conforming spatial

refinement under smooth-solution and initial-projection hypotheses, and whose fixed-mesh interacting kinetic metric is complete, with a unique self-adjoint closure of the declared Laplace–Beltrami Hamiltonian and a nonzero neutral Hilbert space; bounded self-reading software patches execute the same charged model. The geometry layer constructs a $(1+3)$ target from an independent real axis and the rank-three carrier, then isolates the common physical refinement, stress, entropy and scale premises under which null balance yields the Einstein field equation. These are mathematical and computational realizations on supplied geometry and couplings, with physical identification stated separately.

Operational quantum reconstructions [20, 21, 22], relational and modular accounts of physical facts and time [52, 53], thermodynamic routes to gravitational dynamics [63, 64], and compact-group reconstruction [71, 72] are the principal comparison points. Born and Lüders rules, Tsirelson's inequality, conformal–Lorentz isomorphisms, Tannaka reconstruction, and the familiar exterior representation of one generation are used as established ingredients. The originality asserted here is limited to the observer-fiber normalizer with sharp refinement control, the transaction contract yielding the local diamond, the A1–A2 forcing theorem with its fixed-space exclusion, the conditional four-band witness and exhaustive exterior selection, the nine-direction null design with exact constants, the frozen coefficient-linked primitive-port prediction and the carrier-class band containing it, the finite effect-valuation representation with its web countermodels and phase-lift boundary, the converse ladder for the generator of a supplied continuous flow, the kernel-and-reference-fixed log-transition action with its non-identifiability theorem, the two declared-family selections of the internal-energy slope with the hopping-charge gauge and conservation equivalence, the oriented-face compact-family discriminator and the F/G dimension-drop theorem, the exact local face-curvature action with its scalar Coulomb sector, the slot-interface Tsirelson saturation with its record-diagonal boundary, the unique deep-regime enclosed-mass law, the capacity-to-equation-of-state map with its scoring asymmetry, and the identification of the continuum Yang–Mills gap with the limiting repair gap under named certificates.

1.2 Reproduced structures

Table 2 lists the structures of observed physics that the reconstruction returns, with the class of each result in the vocabulary of Table 1 and the section that proves or states it. The table carries no claim beyond the class in its third column: a finite theorem is a theorem about the declared finite objects, a reconstruction implication carries the physical antecedents displayed with it, and a diagnostic consumes measured input and predicts nothing.

Table 2: Reproduced structures of observed physics, the result that returns each one, its claim class, and its location.

Structure	**What the reconstruction returns**	**Class**	**Where**
Probability weights on outcomes	Every nonnegative normalized valuation additive on coexisting effects is the trace against a unique density operator, in every finite dimension, with no continuity axiom	finite theorem	§4
State update after an outcome	A declared Lüders instrument has the exact channel and conditioning identities; a swap-twisted instrument has the same effects, so the effect table does not select the update; invariance on the outcome-certain states does	finite theorem	§4

Table 2 continued

Structure	What the reconstruction returns	Class	Where
The correlation ceiling $2\sqrt{2}$	Tsirelson bound on commuting record subalgebras, attained exactly on a declared slot-split entangled witness, with record-diagonal states confined to 2	finite theorem	§4
Impossibility of copying an unknown state	One ancilla-free isometry copying two sharp states from a common blank forces their overlap to 0 or 1	finite theorem	§4
A three-dimensional position carrier	The normalized twelve-port repair response has an intrinsic rank-three Gram quotient; the signed record module $\mathbb{Z}^6$ and the seam current module D_6 are dense in one abstract three-dimensional Euclidean carrier	finite theorem	§2
Lorentz kinematics and the velocity space	$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ with H^3 the space of future unit timelike directions, on the supplied support-visible cap–BW branch	conditional scaling theorem	§6
A source-derived $1+3$ causal/Lorentz precursor	The independent real axis and exact rank-three source Gram quotient define a four-dimensional ambient carrier with Lorentz inertia $(1,3)$ and the exact S^2/future-null-ray boundary; the authenticated parent order supplies the exact longest-parent-chain height, used only as the temporal coordinate of an event placement after scaling and not as a dimension estimator; a physical manifold on the direct event-order route requires the separately stated source-causal continuum certificate	finite theorem plus reconstruction implication	§6
The relativistic mass shell	$E^2 = m^2 + \lvert p\rvert^2$ for the declared-mass momentum, rest characterization $E = m$ exactly at vanishing spatial coordinates, invariance in every oriented chart, and the massless branch	finite theorem	§9
Internal energy in inertia	For a nonzero ledger, frame covariance selects slope one in the declared momentum family and, independently, refinement and the declared proper-time principle select slope one in the length-action family, giving inertial coefficient $m + E$; the ledger is additive over product references with an exact binding defect	finite theorem	§9
A nonlinear matter continuum	First-order trajectory convergence in $H^1 \times L^2$ for the real quartic-wave sector of the supplied charged action, on uniform conforming refinements with smooth Neumann data and Ritz initialization	conditional analytic theorem	§9.6
The arrow of time	Every finite stochastic repair kernel preserving a supplied faithful reference contracts relative entropy; the deterministic strict-descent normalizer need not, as an exact counterexample shows	conditional finite theorem	§11
The four laws of thermodynamics	One exact conditional package with five typed source and physical receipts, including the transition-side reading of the third axiom; it is not a source-realized theorem of the three axioms	conditional finite theorem package	§11

Table 2 continued

Structure	What the reconstruction returns	Class	Where
Linear transport coefficients	Symmetric positive-semidefinite finite Green–Kubo matrix with an exact cutoff remainder	finite theorem	§11
Entropy that scales with a boundary	Invariant collar decomposition with the one-sided edge term $\sum_\alpha p_\alpha \log d_\alpha$; on the sparse read family the crossing-read functional Γ-converges to Euclidean perimeter, with balls as the limiting minimizers	finite theorem; conditional analytic theorem	§5, §6
The inverse-square law	Spherically symmetric flux at constant shell charge falls as r^{-2}, the exponent being the carrier dimension minus one, with the dimension supplied by the completion theorem	finite theorem	§7
Coulomb and Maxwell structure	Symmetric rational Green matrix with a unique minimal-energy neutral solution, and a local twenty-face curvature action gauge invariant exactly for conserved seam currents	finite theorem	§9
The Einstein field equation with Λ	Tensor completion of a realized null balance under Ward conservation, the Bianchi identity, and one scale identification	reconstruction implication	§7
The Standard Model gauge Lie type	Complete A1 response and endogenous A2 transport force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ on twelve ports, with no ambient gauge group among the premises	finite theorem	§8
Common kernel and maximal faithful image on the declared matter module	Common $\mathbb{Z}_6$ kernel on the declared matter tensors, so their maximal faithful image is $S(U(3) \times U(2))$; this does not select the physical global form	conditional finite algebraic theorem	§8
One anomaly-free generation of fifteen states	Exhaustive scan of the declared exterior menu leaves exactly one charge-conjugate pair of rank-fifteen chiral anomaly-free masks	finite theorem	§8
Absence of the minimal grand-unified exchange channel	The product adjoint $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$ contains no X/Y generator	finite theorem	§8
Massless photon, gluon, and graviton	Three conditional structural statements on their declared branches, each with a zero hard mass parameter: two photon modes, two transverse modes per Yang–Mills generator, and two transverse–traceless gravitational modes	reconstruction implication	§8.4
A positive Yang–Mills mass gap	The continuum Hamiltonian gap identified with the limiting repair gap only on the support-visible compact-gauge branch carrying the finite-rate, transfer, and reconstruction certificates	reconstruction implication	§8.5
The Koide relation among charged leptons	Positive-chamber circulant identity with $Q = 2/3$ exactly at $\rho/a = 1/\sqrt{2}$, plus a custody-bound tau window	finite theorem; fixed test	§10
Charged-lepton mass magnitudes	Outward-rounded enclosures of logarithmic half-width 1.732% containing the measured triple	diagnostic	§10

Table 2 continued

Structure	What the reconstruction returns	Class	Where
The fine-structure constant	Interval-certified fixed points of the screen-grain closure loop, 136.994835 for the bare source map and 137.035660 for the self-consistent gauge-width map, against the measured 137.035999; the gauge width closes 99.2 percent of the gap	diagnostic	§10.1
The cosmological constant	Capacity closure candidates 3.2921×10^{122} and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck chain	diagnostic	§10.1
Flat rotation curves and the baryonic Tully–Fisher relation	Given declared deep-regime scale covariance and quadrature source composition, the unique law is $M_A(r) = r\sqrt{M_b a_0/G}$ and gives $v^4 = GM_b a_0$; neither premise nor a_0 is source-derived	conditional characterization theorem	§11.1
A gravitating non-luminous sector	Modular-charge sourcing classifies any geometric excess over the luminous term, with the anomalous collar energy as a candidate carrier	reconstruction implication	§11.1
The dark-energy equation of state	Under the declared inverse-density reading $\rho_{\mathrm{DE}} = \kappa/N$, fixed record capacity gives the exact point $(w_0, w_a) = (-1, 0)$, and monotone nondecreasing capacity gives $w \geq -1$ at every epoch	reconstruction implication	§11.1
The de Sitter transfer sign	Exact capacity identity with $\Delta S_{\mathrm{gen}}^{\max} = \log(1-f) < 0$ and a one-sided boundary maximum	finite theorem	§10
Isotropy of propagation to angular rank five	The first symmetry-allowed anisotropic rank of the carrier class is six, with two frozen rays inside one exact band	finite theorem; frozen prediction	§10.4
An interacting quantum state space	Complete reduced kinetic metric on $\mathbb{R}^{30} \times \mathbb{C}^{13}$, unique self-adjoint closure for the declared Laplace–Beltrami Hamiltonian, and invariant neutral states	conditional analytic theorem	§9.6

Every row carries the premises stated at its own theorem, and the physical realization maps that separate a finite implication from a claim about nature are collected in Section 11.

1.3 The standing unsupplied maps

One physical realization must carry compatible events, fields, action, states and readouts across refinement. Its five obligations are:

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, with a physical length, cofinal overlap gluing between carriers, and a refinement limit.

(M2) Clock and energy calibration. An identification of the internal step with a laboratory clock and of the modular ledger with a laboratory energy, in measured units; configuration-derived model duration does not select laboratory time or its units.

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) A source-selected action. An action produced by the source rather than declared, with its field sector, coupled interaction, relative normalizations, and operational readout.

A finite theorem that consumes none of these stands on its own domain. A statement about nature consumes some of them, and each boundary block below names which. No result in this paper discharges any of the five.

1.4 The three axioms

The model uses the following three mathematical postulates.

A1 (carrier federation and spherical support). In plain language, the model assigns local state, readback, records, repair moves, and checkpoints to a federation of observer-patch carriers. Each local carrier has an oriented twelve-port boundary. Typed overlaps form the federation nerve, which maps to a separate oriented spherical support. Concisely, for each regulator r there is a typed object

$$\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$$

whose carrier boundary $K_{r,i} = (P_{r,i}, E_{r,i}, F_{r,i}, o_{r,i})$ has $|P| = 12$, $|E| = 30$, $|F| = 20$, and the oriented icosahedral incidence relations. The bridge $b_r : N_r \to S_r$ carries a designated oriented two-cycle to the fundamental class of the spherical support, and the typed maps commute with refinement. For each complete carrier, put $V_{r,i} = \mathbb{R}^{P_{r,i}}$. A1 supplies a finite-dimensional unitary space $H_{r,i}$ and an injective real-linear response derivative

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i})$$

whose image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators. The primitive port probes span $V_{r,i}$, their ordered compositions generate every accepted infinitesimal reversible response, and no public response direction is omitted. The pairing

$$\langle v, w \rangle_D = -\operatorname{Tr}(D_{r,i}(v) D_{r,i}(w))$$

is positive definite. The response data and their completeness commute with refinement, and

$$G^0_{D,r,i} = \langle \exp(t D_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0$$

is the connected response group. A1 constrains the finite carrier, federation, operational interfaces, spherical-support tower, and complete public reversible-response tangent. It does not imply semantic agreement, repair termination, confluence, a particular matrix response, an inverse-port law, a compact Lie type, a global group, or a laboratory identification [4].

A2 (observer agreement). In plain language, observers assign the same operational meaning to accepted data on their shared boundary. Every proper recharting of a complete carrier is implemented by their own reversible overlap response. Formally, the interpretation functor

$$\mathcal{J}_r : \mathsf{Data}_r \longrightarrow \mathsf{Meaning}_r$$

is natural under every visible restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map. On an overlap O,

$$\mathcal{J}_O(\mathrm{res}_{P\to O}\, d_P) = \mathcal{J}_O(\tau_{Q\to P}\, \mathrm{res}_{Q\to O}\, d_Q).$$

The accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, let $\mathrm{Hol}_r(o)$ be its closed overlap paths. A2 requires the port projection

$$\Pi_{r,i} : \mathrm{Hol}_r(o) \longrightarrow \mathrm{Aut}^+(K_{r,i})$$

to be surjective. For every proper carrier automorphism $a \in \mathrm{Aut}^+(K_{r,i})$, A2 supplies a closed path γ_a and a projective unitary implementer $[U_a]$ satisfying

$$\Pi_{r,i}(\gamma_a) = a, \qquad \mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a\cdot v).$$

The implementer is endogenous to the same response:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \qquad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

The centralizer factor acts trivially on $\mathfrak{g}_{r,i}$. A2 constrains operational meaning on accepted shared data and makes proper carrier rechartings inner on the complete port-response algebra. It does not imply global state extension, termination, confluence, unique normal forms, durable records, a global compact group, or a laboratory current [4].

A3 (maximum randomness under agreement constraints). In plain language, the selected state is least informative relative to the declared reference after every observer-visible constraint has been imposed. At finite regulator r, let $\mathcal{K}_r$ be the nonempty convex set of compatible local state families and let

$$\mathcal{D}_r(\rho\|\tau_r) = \sum_{P\in\mathcal{G}_r} w_{r,P} D(\rho_{r,P}\|\tau_{r,P}), \qquad w_{r,P} > 0.$$

The observer cover is state-determining on $\mathcal{K}_r$, and

$$\rho_r = \operatorname*{arg\,min}_{\rho\in\mathcal{K}_r} \mathcal{D}_r(\rho\|\tau_r).$$

A3 constrains state selection inside one A1-fixed feasible space. It does not select a field list, repair law, response map, particle multiplicity, or continuum limit [4].

None of the axioms contains a gauge group, a particle list, a spacetime dimension, or a recovery law. A1 does fix a two-dimensional spherical support and hence the Lorentz kinematic type used below. The results below state which structures follow from these axioms, which require added premises, and which realization maps are consumed as premises.

1.5 The self-referential closure hypothesis

The closure hypothesis is the overarching structural premise. The universe is modeled as a timeless, closed, self-referential mathematical structure in which the simulating description and the simulated system are one object, and the complete inhabited history is a fixed point $T(\mathfrak{U}) = \mathfrak{U}$ of a universe-level closure operator. No primitive or preferred global clock is assumed or derived: repair order is a partial order on commits, an observer history is an ordered chain of completed records, and

that order is invariant under strictly increasing regradings, so affine time and proper time require separate calibration and event-geometry receipts. The twelve-port carrier belongs to the fixed-point solution inhabited by the observers, and A1 is their recovered description of its quotient-visible specification. Closure requires the hardware present in the structure, the specification inferred from public records, and the realization constructed later in observer record order to return the same invariant quantities. The mathematical theorems consume that specification as a premise; universe-level closure is the separately stated hypothesis and supplies no finite repair law, current derivative, physical kinetic operator, or existence and uniqueness proof for the cosmic fixed point. The history's local events and their observable and interventional dependencies constitute its proposed intrinsic computation. Fixedness of the complete history permits changing states within it; it does not identify a last instantaneous state with the universe.

No temporal paradox arises. The equation relates parts of one timeless object, with no global time coordinate along which information travels backward. Repair descent is a well-founded order on authorized state updates used to prove termination and compare schedules; it is neither an observer's experienced time nor a universe-level clock, so observers can recover the specification and later construct it in their own record order while both stages belong to the same global fixed point. On this reading A1–A3 are the finite consistency conditions under which the loop closes, and the A2 endogeneity clause is the hypothesis's local expression inside the axioms: every proper recharting is implemented from inside the response. Two finite results give the hypothesis exact content on declared branches. The packet-quotient closure theorem proves, on the declared finite consensus branch, that the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms, with no habitat-level extension [7]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated universe-level observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves nonidentifiability for the bounded completion class defined by base agreement, positivity, and the carrier bound; universal all-rung membership in a complete A1–A3 capacity-source contract is unproved, so direct N is not evaluable on that incomplete antecedent [10]. A positive result requires a complete antecedent and a proof of one physical zero.

Its quantitative candidates, the screen-grain equation and the capacity coordinate, are developed in Section 10.1, where existence, uniqueness, and stability of the physical fixed point are separate determinacy tests.

1.6 How the constraints work together

Physical structure is selected at the intersections of constraints. The carrier incidence fixes a finite response space with its symmetry action and invariant subspaces, and that geometry by itself admits many equivariant response laws; A1 restricts the menu to faithful, complete, compact, commutator-closed reversible responses, and A2 requires every proper recharting visible through overlap agreement to act internally on that same response, so for the twelve-port carrier the one-dimensional fixed space excludes the sole centreless compact alternative and fixes the local Standard Model gauge algebra. Repair dynamics occupy a distinct part of the architecture: a named repair law produces a schedule-independent public record only when its termination and local-diamond premises hold, and the gauge theorem imports no such law. Carrier counting and algebraic traces supply finite candidate measures, agreement and repair determine whether they descend to the public quotient, and a physical realization map must then preserve them as spacetime or laboratory measures. This dependency discipline separates forced structure from formal resemblance.

1.7 The theorem chain

The results form branches with different premises. Finite carrier and convergent repair give the consensus normal form; signed port records with the normalized repair response give the abstract continuous three-dimensional Euclidean carrier; the algebra-state reading of the completed record gives the finite quantum record identities; A1 response with A2 transport gives the Standard Model local Lie type, and a displayed matter representation the conditional $\mathbb{Z}_6$ matter image; oriented spherical support gives Lorentz observer-velocity kinematics; and authenticated read-after-write provenance gives the finite informational causal poset with its canonical source height, which with the rank-three carrier, real axis and celestial S^2 becomes the populated finite $(1+3)$ Lorentzian precursor with exact order–cone embedding. These rows are finite or exact results under their displayed premises, the matter-image row a conditional finite algebraic theorem. The reconstruction implications follow: physical event and link attachment with compatible refinement, calibrated count–volume density, independent dimension, manifoldlikeness, topology and uniqueness give the effective four-dimensional time-oriented Lorentzian event manifold; same-family tensor-curvature convergence or continuum small-ball/null balance, with stress, Ward/Bianchi, coupling and scale, give $G_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}$; deep-regime scale covariance with quadrature composition gives $v^4 = GM_b a_0$; and repair relaxation with reconstruction certificates gives the positive Yang–Mills mass gap. A common modular–stress–entropy tower is the proposed route to the null-balance antecedent of the Einstein row, and no physical realization of that tower is supplied. Sections 2 through 10 develop these branches, Section 11 records the physical interpretation maps, and Sections 12 to 14 give the verification stack, the falsification architecture and the comparison with neighboring approaches.

2 The primitive finite architecture

Definition 2.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 2.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

The carrier realization used throughout is the twelve-port icosahedral architecture of A1: each carrier's boundary packet is combinatorially the oriented icosahedron boundary (P, E, F, o) with $|P| = 12$, $|E| = 30$, $|F| = 20$; carriers federate through typed seam algebras with coherent triple-overlap cocycles into a nerve carrying a degree-one bridge to the oriented spherical support. Under the closure hypothesis, this is the quotient-visible source architecture recovered by inhabitants of the fixed point. Every result below displays its dependence on that architecture.

Two exact features of this architecture enter the later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports; the real port-coefficient space has the character decomposition

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \tag{1}$$

an exact multiplicity-free decomposition obtained by standard finite-group character theory [58] and checked by certified projectors. The two triplet bands are machine-checked at the level of the

projectors themselves. On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$ with eigenvalues $2\varphi^2$ and $2/\varphi^2$, each is the entrywise Galois conjugate of the other, their characters take the values $3, -1, 0$ off the order-five classes and φ or $1-\varphi$ on the twenty-four order-five elements, and a Burnside span certificate makes each piece an irreducible real representation of the listed automorphism group. Irreducibility survives extension of scalars: the nine listed automorphisms span the complex endomorphism algebra with the same inverse certificates, the two complexified pieces are inequivalent because the traces on the order-five row are $1-\varphi$ and φ, and the real endomorphism algebra of each piece is the real scalars, so both are of real type (`Lean/Screen/GoldenSectorComplexIrreducibility.lean`).

An independent exact quaternion construction provides a separate character theory control. Direct enumeration of the standard H_4 coordinates over $\mathbb{Q}(\sqrt{5})$ and closure from two generators recover the same 120-element binary-icosahedral carrier, the complete irreducible system and affine E_8 fusion graph. A faithful doublet has character field $\mathbb{Q}(\sqrt{5})$; its Galois-conjugate doublet has the same graph type. The graph therefore does not choose φ over $1-\varphi$. This executable reconstruction supplies no physical embedding selector or explicit isomorphism to the separately enumerated $\mathrm{SL}(2,\mathbb{F}_5)$ carrier [8].

The local oriented seed and the level-zero support triangulation have the same face table. Their sixty orientation-preserving identifications form an A_5 torsor and each sends the fundamental cycle exactly to the designated support section. The fixed source decoration singles out one vertex and leaves a C_5 stabilizer, giving twelve orbits under that fixed-mark action; these are not twelve physical alternatives. A source-selected attachment, a compatible local refinement tower and a common oriented-sphere realization are required before the degrees one and seven of the two Galois frames support a source-bound homotopy comparison. Physical identification is a separate requirement [8]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port; a conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(N) = \sum_i N_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates by a finite theorem. Divisibility of total load by twelve is a necessary consensus condition and is not sufficient by itself. For the certificate's source-generated full-pile packet, an explicit eighteen-move schedule reaches consensus.

The same incidence fixes the dimension of the carrier's own position readback, which is the reason the number three occurs in this framework without being declared.

Theorem 2.3 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $T = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

Boundary 2.4. The completion is taken after slow-response normalization and the infinite-step limit; completing at finite step count gives a different object, since the kernel has rank eleven on the centered space and rank six on the signed antipodal sector. The metric here is the response-selected Gram pullback and is distinct from the usual six-dimensional lattice metric on D_6. The nonlinear repair kernel does not descend through the signed quotient even though its conditional mean does, so working state and ordered history form a separate fiber. Thirteen finite covariance and isometry

table identities used by the proper-action packet are discharged in Lean with `native_decide`; those subreceipts therefore trust the native compiler and runtime in addition to Lean's kernel and are not kernel-only proofs. Reading the completion as physical carrier position consumes the source readback and distinguishability premises; scale refinement, cofinal overlap gluing, physical space, physical scale, and field attachment are not constructed [1, 8].

2.1 Hardware and protocol: one substrate, typed dynamics, several readouts

The finite substrate and the operations acting on it are distinct parts of the model, and every physical conclusion requires a specified operation and its own realization map. The *substrate and reversible-response tangent* are the content of A1: a federation of finite carriers, each with the oriented twelve-port icosahedral boundary packet, joined through typed seam algebras with coherent triple overlaps into a nerve that bridges to the oriented spherical support, together with each carrier's complete reversible response space [8]. This layer fixes what exists at each regulator and selects no repair schedule, convergent repair law or propagation operator. The *agreement and state-selection protocol* combines A2 and A3 with named branch dynamics: patches write and retract integer record events, read back port sums, compare induced records on shared boundaries, and interpret accepted data through the natural functor of A2; a named repair law moves oriented mismatches under quadratic descent, and A3 selects the residual state by constrained information projection once the agreement constraints are fixed [7]. Under the closure hypothesis the substrate is also the specification recovered by observers inside the model and constructed by them within their emergent future.

Distinct readouts of this one protocol-on-hardware system yield the distinct pillars of observed physics. Repair to a fixed point under quadratic descent and the transactional diamond gives the schedule-independent public record of Section 3; the algebra-state reading of that record surface gives the quantum identities of Section 4; overlap transport around closed paths gives the central defects and edge-center entropy of Section 5; authenticated semantic commits give the event carrier and order, and with the real axis, source rank three and celestial cone data the finite $(1 + 3)$ model of Section 6, whose conditional Einstein limit is Section 7; endogenous recharting of the complete port response gives the gauge type and conditional matter image of Section 8; record readback, the modular ledger and the realized path law give the position, energy and action of Section 9; capacity accounting and closure readback give the quantitative surfaces of Section 10, and homogeneous propagation on a selected carrier action gives the coefficient rays of Section 10.4. The branches consume one substrate under typed operations, so the same completed records that define measurement feed the event geometry, and the same overlap transport that defines agreement feeds the gauge forcing. The universe claim of the program is the conjunction: a physical carrier that realizes the hardware and runs the protocol produces public records whose readouts include quantum probability, Lorentzian causal structure, Einstein dynamics, and the Standard Model gauge and matter skeleton, with the maps of Section 1.3 as additional premises.

3 Observable normal forms and finite consensus

A completed finite configuration x is a fixed point of the named repair-and-agreement map:

$$\mathcal{R}_{\mathrm{cons}}(x) = x, \tag{2}$$

where $\mathcal{R}_{\mathrm{cons}}$ is the composite finite dynamics. This map is distinct from the universe-level closure operator T of Section 1.5. Fixed-point existence, observable determination, and schedule independence are separate questions. The following framework isolates them [5].

3.1 Exact observable fibers

Definition 3.1 (Observable quotient system)**.** An observable quotient system is a tuple

$$\mathfrak{S} = (Q, C, \mathcal{B}, B),$$

where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 3.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $N : Q \to C \sqcup \{\bot\}$ *that preserves* B *on its* C*-valued outputs, fixes every point of* C*, is constant on each* B*-fiber, and returns* $\bot$ *exactly on fibers with no consistent extension;*

(iii) $B|_C : C \to B(C)$ *is a bijection.*

If $B(Q) \subseteq B(C)$*, the map is a total idempotent retraction* $Q \to C$*.*

Proof. Injectivity makes every nonempty C_b a singleton, which defines N and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = N(c) = N(c') = c'$. The third statement is the restriction of the first to the image $B(C)$. □

This normalizer is defined by observable fibers, independent of a chosen repair schedule. A repair relation implements it only when the relation preserves B, reaches the consistent set, and satisfies an appropriate liveness condition. Confluence from one source does not by itself prove determination across distinct sources with the same protected record.

For a separately declared finite scheduler, protected observations admit four nested first-hit layers, nonempty consistent fiber, positive reach from some active source, almost-sure reach from every active source, and a single first-hit endpoint class modulo the declared silent equivalence; their complementary cuts partition the empty-fiber, inaccessible, liveness and selection obstructions, and an exact finite-state morphism transports the whole profile [5]. For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable state a normal form, work conservation bounds the normal horizon by the initial mismatch count, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law: a scheduler can delay the first genuine repair for any prescribed number of attempts, so mismatch rank alone gives no attempt horizon, while a bounded-waste upper horizon of $(q + 1)$ times the initial mismatch count holds when every reducible record sees a genuine change among the next $q+1$ attempts, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts. The compiler's single-consumer structure bounds every accepted path by $n(n+1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained

by a fanout chain. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [5, 7].

The precedence layer beneath these results is source-derived. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the register versions its acceptance depends on, and the informational order is the transitive closure of authenticated read-from parenthood: a direct edge requires the child to certify the resource, the parent to write it, the child's pre-commit snapshot to name that parent as writer, and the pre-commit value to equal the parent's post-commit value. The supplied dependence set must lie inside the read set; its soundness as the set of genuine dependencies is a producer and verifier obligation rather than kernel-derived analysis. The result is a strict order, least among strict transitive extensions of the direct edges, and an authenticated analogue of happened-before in distributed computing [46]. Static mismatch scores are writer-blind, so a standing residual determines no causal arrow; under fresh-identifier, duplicate-free and visible-support hypotheses a hidden commit changes no visible mismatch and acquires no outgoing parent edge to a visibly supported child, and one adjacent independent swap preserves the authenticated relation. Record or program order contributes only when a later commit consumes the earlier record.

Lean packages reflexive generated ancestry as a partial order with finite intervals at each finite cutoff, so as an abstract finite poset the object satisfies the order-theoretic causal-set axioms (`generatedBeforeEq_isPartialOrder`, `finiteCausalSetAxioms`). Causal set theory starts from a locally finite order read as proto-causality [31, 32]; here finiteness gives local finiteness, and read-from dependence is informational rather than physical. A faithful-embedding interpretation additionally requires an order embedding, manifoldlikeness and dimension, and the approximately Poisson count-density condition [33, 34, 35]; on the continuum side, causal order determines conformal geometry only under distinguishing or causality hypotheses [44, 45], a volume datum is needed for the conformal factor, and the Lorentz-invariant continuum approximation uses Poisson sprinkling rather than a regular lattice [43]. The commit log supplies no calibrated map from event counts to spacetime volume, and no faithful embedding, manifoldlikeness, volume law, continuum existence or uniqueness theorem follows from the finite provenance order alone.

Theorem 3.3 (Finite causal-order compiler)**.** *Let E be finite and let $R \subset E \times E$ be a decidable irreflexive transitive relation. There is an abstract authenticated semantic log L_R on E whose direct semantic-parent relation and generated strict precedence are both exactly R:*

$$L_R.\mathsf{ParentEdge}(e, f) \iff R(e, f), \qquad L_R.\mathsf{GeneratedBefore}(e, f) \iff R(e, f).$$

No rank is supplied: the compiler uses the number of strict predecessors, which increases strictly along R. In this generic construction every supplied strict predecessor is an authenticated semantic parent: it represents R itself rather than its Hasse reduction alone.

Proof. Give each event its own register; e writes it and f certifies it exactly when $R(e, f)$, with unit values and the writer stamp authenticating identity. Irreflexivity and transitivity make predecessor sets grow strictly along R, so predecessor cardinality is a rank; authenticated parenthood is R by construction and its generated closure is R by transitivity. □

Chainlike captured histories are therefore not forced by the authenticated-provenance grammar: every finite causal set is an abstract semantic log, and Theorem 6.2 gives it a separated one-way realization in the exact source $(1 + 3)$ carrier. The relation R is supplied, and no OPH dynamics, physical event identification, calibrated density law or manifold limit is selected by this expressivity result (`Lean/ObserverPatchHolography/Provenance/FiniteCausetCompiler.lean`).

The ambient carrier's dimension and signature are not left to a chart receipt. Authenticated provenance fixes the events and their generated order; independently, the exact source Gram quotient is a positive rank-three carrier whose direct sum with a real axis has Lorentz inertia $(1, 3)$, with the source S^2 as its null boundary. Section 6 states the placement theorems that join the two structures, the continuum certificate a physical manifold on the direct event-order route requires, and the finite Einstein-shape identity that exact null tomography returns on the same event type.

A supplied scalar action on prepared golden addresses supports classical and Fock-space waves. Its massive Dirichlet detector limit survives spatial refinement and symmetric local steps; at $q = 233$, compact-detector error is smaller than the induced signal on a nonzero window. On a 64-site action an authenticated classical history closes a feedback loop, detector probes perturbing the field, retained records restoring it and later updates consuming the restored values, with separated intervention response above 0.01576 throughout a nonzero model-time interval against a total declared error below 0.01364; a quantum instrument on the same action prepares its coherent intervention by a local force pulse within trace distance 9.389×10^{-6} of the target [11]. Axes, boundary, addresses, action, vacuum, controls and model time are supplied; physical clock and count-volume attachment remain separate.

The spacetime derivation develops the finite read algebra of this substrate and its boundaries [11]. In the finite Schrödinger representation of a scalar action, a region's canonical algebra is generated by its own fields and exterior linear fields exactly when the exterior fields cover the row space of the coupling block B, so $\operatorname{rank} B$ exterior fields are necessary and sufficient; on connected support one receiver's N retained exact samples recover all N initial scalar values with at most $N(N-1)/2$ supported means; for a declared public record map and deadline the unique constrained information projection is the reference conditioned on publishing histories, and with the deadline removed the continuation probability is the least harmonic majorant of the publication indicator, giving the exact eventual-publication probability $p_1/(p_0 + p_1)$ for two unknowns on an isolated path. For the finite A3 objective itself, with faithful local references and positive cover weights, every minimizing local density has the union of the feasible supports, and a feasible witness with failure probability $b > 0$ gives the floor $\exp[-(F(\sigma)/w_P + H(b))/b]$ on its selected probability, with F the weighted relative-entropy objective and H binary entropy. The same source carries the non-identifiability boundaries: the registered all-port capture has normalized read rank 23 for every even $N \geq 4$ and admits positive preparations agreeing on every local payload while disagreeing on remote records, every radius $Lq^{-\alpha}$ with $0 < \alpha < 1$ and uniform cube populations pass the same limit tests, and no finite read theorem selects the declared read law, population or radius from the complete source grammar. These are access counts and finite selection identities, not physical precision, routing cost or clock statements.

3.2 Approximate outputs, refinement, and cost

Three further results control the normalizer where exactness fails, with proofs in the component paper [5]. Let (Q, d_Q) be finite, $(\mathcal{B}, d_{\mathcal{B}})$ metric, and $C = \Phi^{-1}(0)$ nonempty for a nonnegative consistency residual Φ; let $\eta_\Phi(t) = \max\{\operatorname{dist}_Q(x, C) : \Phi(x) \leq t\}$ turn residual error into distance from consistency and $\omega_B(r) = \max\{d_Q(c, c') : c, c' \in C,\ d_{\mathcal{B}}(Bc, Bc') \leq r\}$ measure how well protected observations distinguish consistent outputs. For L_B-Lipschitz B, any two outputs with $\Phi(x) \leq \delta_x$, $\Phi(y) \leq \delta_y$ and $d_{\mathcal{B}}(Bx, By) \leq \varepsilon$ satisfy

$$d_Q(x, y) \leq \eta_\Phi(\delta_x) + \eta_\Phi(\delta_y) + \omega_B(\varepsilon + L_B[\eta_\Phi(\delta_x) + \eta_\Phi(\delta_y)]) \,, \tag{3}$$

exact in the ω_B term at zero residual and with the coefficient one on each residual-distance term optimal among finite metric systems. Along a refinement tower with restriction maps $\rho_{m,n}$ and

level normalizers N_n, the naturality defect accumulates as $d_n(\rho_{m,n}N_m q, N_n\rho_{m,n}q) \le \sum_{j=n}^{m-1} K_{j,n}a_j$ with $K_{j,n} = \mathrm{Lip}(\rho_{j,n})$ and a_j the one-step defect, so nonexpansive restrictions with a summable defect sequence give asymptotic naturality. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, deciding whether $B_S|_C$ is injective is coNP-complete even when all but one variable are observed, and deciding whether every protected collar value has some consistent interior extension is Π_2^P-complete.

3.3 Repair realization on the twelve-port carrier

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 3.4 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak

$$t \longleftarrow s \longrightarrow u$$

has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint in the sense $W_\tau \cap (R_\sigma \cup W_\sigma) = W_\sigma \cap (R_\tau \cup W_\tau) = \varnothing$, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 3.5 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies Proposition 3.4, the exact quadratic descent of Section 2, and repair completeness, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 3.4 makes it locally confluent, so Newman's lemma [51] makes it confluent. A terminating confluent relation has a unique normal form below each source. Repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Boundary 3.6. Atomic commits, disjoint write sets, or termination alone do not imply the local diamond. Countermodels omitting semantic dependency closure, canonical aggregation, or the surviving component rule are retained with the finite receipt. The reference engine exhaustively checks its states, peaks, payload hashes, protected records, and descent comparisons. The theorem is finite and asserts no continuum limit.

The declared class also has gauge compatibility, idempotence of the completed dynamics, and refinement compatibility where proved; the public object is the protected quotient record, not an intermediate configuration.

The public OPH-FPE run of this law uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighbouring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive, so uniform scheduling gives the repair mean $T = I - L/60$ of Theorem 2.3 on every carrier. Under the integer law, sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising $V = \sum_i N_i^2$, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals; under the real seam-mean law the component mean is the unique fixed point of the glued federation. The per-carrier normalized response kernel read from the run equals $4P_{\mathrm{slow}}$ to 10^{-12} on isolated carriers, while under the declared gluing the glued-port pattern replaces the slow-band share at large depth, a property of the gluing convention; cofinal gluing is not constructed. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed, and recomputes every hash [2]. The same law at the level-ten rung, 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at the balanced class in 84 to 100 sweeps with the one quotient hash fixed by the loads. Across the tested rungs from six to ten, one initial load per rung gives an empirical excess-energy contraction near 0.82 per sweep and a fitted settling time $4.3 \log_2 N - 9$. These finite fits prove no all-size mixing or termination-time bound. The recorded strict-descent count is about 1.2555 times half the initial ℓ_1 deviation from the uniform reading; it excludes energy-preserving swaps and is not transported mass. The isolated per-carrier response kernels agree across rungs to six digits.

The first-settling configuration retains spatial information that its public component multiset does not encode. A controlled readback of all sixteen schedules at each of levels eight through ten separates common input memory from schedule variance using equal-count ancestral cell groups: at level ten the initial-to-terminal amplitude responses at group sizes 1,024, 4,096 and 65,536 are 0.90758, 0.95232 and 0.98983, the conditional schedule-noise fractions 1.241%, 0.594% and 0.143%, and at the largest group the terminal variance is 11.048 times the stationary fixed-total balanced null. The schedules share one initial load, the groups carry no derived physical area, and each snapshot ends the first settling sweep; this is finite coarse memory under the declared gluing, without a primordial-spectrum identification, with the executable statistics and input digests retained in `evidence/observer_dynamics_20260925/`. Stationary continuation supplies a separate constructive fluctuation result. For p ports on a connected weighted graph, $0 < r < p$ raised occupancies, fair integer repair and independent edge-attempt clocks, write $z = n - (r/p)\mathbf{1}$, $\Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/p$ and $\kappa = r(p-r)/(p(p-1))$; with graph Laplacian L in attempt-rate units, the uniform fixed-occupancy

law is reversible and

$$\mathbb{E}[z(t)z(0)^{\mathsf{T}}] = \kappa e^{-|t|L/2}\Pi, \qquad T^{-1/2}\int_0^T Bz(t)\,dt \implies \mathcal{N}(0, 4\kappa BL^{+}B^{\mathsf{T}})$$

for a fixed linear readout B, with L^{+} the mean-zero inverse. The finite-state martingale central limit theorem applies at fixed graph, with no uniform refinement limit asserted: this constructs a history source from local port repair without adding Gaussian site noise, its occupancy law, elapsed-time convention and readout are specified inputs, and it does not identify this source with a geometric volume perturbation or determine the cosmic microwave background amplitude, tilt or expansion history. The screen-microphysics analysis gives its proof, finite-window law and locally sampled event-record correction.

At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations. Tetrahedral and octahedral controls recover their distinct specifications; endpoint overwrite loses conservation and schedule-independent termination.

Complete federation gluing is not identified by the retained level-three routing logs, 22,818 events on 86 of 7,650 seams over 1,280 carriers: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class; neither carrier replay nor this finite obstruction supplies universe-level closure or a physical realization.

4 Finite record algebra and quantum identities

Once a compare, write, and verify slice has completed, its accessible events are represented in a finite-dimensional $*$-algebra with a normalized state. This algebra-state representation is an explicit input at this stage. The consensus theorem identifies which record is public; it does not derive the classification of finite-dimensional C^*-algebras or the trace pairing.

Theorem 4.1 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E\,\rho\,P_E}{\operatorname{Tr}(\rho P_E)} \tag{4}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, define

$$S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0+B_1) + A_1(B_0-B_1)\}]\,.$$

Then

$$|S_{\mathrm{CHSH}}| \le 2\sqrt{2}. \tag{5}$$

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$; for a complete orthogonal partition (P_k), linearity and $\sum_k P_k = I$ give $\sum_k \operatorname{Tr}(\rho P_k) = 1$; and for nonzero event weight $P_E\rho P_E/\operatorname{Tr}(\rho P_E)$ is positive, normalized, and supported on $P_E\mathcal{A}P_E$. For $\mathcal{B} = A_0(B_0+B_1) + A_1(B_0-B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$; the two commutator norms are at most two, so $\|\mathcal{B}\| \le 2\sqrt{2}$, and its expectation in ρ gives the bound. □

Theorem 4.2 (Exact Tsirelson saturation and the diagonal boundary)**.** *The bound of Theorem 4.1 is attained exactly on the committed slot-split interface: the normalized maximally entangled state on the four-dimensional product carrier, with the two Pauli projections on the left slot and the two quarter-rotated projections on the right slot, gives* $S_{\mathrm{CHSH}} = 2\sqrt{2}$*, and the four events inhabit the declared finite subsystem split with the identity Kraus family. Conversely, every state diagonal in the record basis, paired with any four diagonal observables valued in* $[-1, 1]$*, obeys the classical bound* $|S_{\mathrm{CHSH}}| \leq 2$*; in particular the counted correlation state of the committed source pair obeys it.*

The two halves bracket one committed record surface: the algebra-state representation attains the full quantum value on a declared entangled witness, while one record-diagonal state with four jointly diagonal bounded readouts stays inside the classical bound. The Bell state and all four settings are real matrices, so the gap is entanglement and noncommutativity rather than the complex phase direction; the saturating state and events are declared library data rather than source-produced preparations.

The first identity is the Born rule on this finite surface [23]; the second is the declared Lüders conditioning instrument [24]. The projectors alone do not select that instrument: an explicit swap-twisted instrument has the same effects and different updates. Kraus form with induced effect P together with invariance on the states certain of P is equivalent to being the Lüders map, so certain-state invariance is what selects the instrument the effect table leaves open; the swap-conjugation map on two dimensions has Kraus form with induced effect the identity and fails that invariance, and the committed repeatability clause is strictly weaker and does not select (`Lean/EventAlgebra/InstrumentSelectionByCertainStates.lean`). Events, preparation, readback, and a source implementation are unbuilt. The third is the Tsirelson bound [26] for the Clauser–Horne–Shimony–Holt combination [25] at fixed cutoff. Compatibility structure for commuting events follows the standard finite operational pattern within the same library.

Two maps built from a partition of pairwise orthogonal projectors (P_i) with $\sum_i P_i = I$ organize conditioning on the same surface: the pinching $\mathcal{P}(X) = \sum_i P_i X P_i$ and the average $\mathcal{A}(X) = \sum_i \frac{\mathrm{Tr}(XP_i)}{\mathrm{Tr}(P_i)} P_i$ are positive, unital, trace-preserving idempotents whose exact ranges are the commutant of the partition and the commutative span of the P_i, so $\mathcal{A}\mathcal{P} = \mathcal{A} = \mathcal{P}\mathcal{A}$, and on a member of nonzero weight the average commutes with Lüders conditioning. These are standard finite-matrix identities related to trace-preserving conditional expectations [27, 28]; the formal development contributes a checked arbitrary-partition interface, exact range and uniqueness statements, and adapters from projection events to the state-level Clauser–Horne–Shimony–Holt bound [6], with no priority claim for Born probability, Lüders conditioning, pinching, or Tsirelson's inequality as identities of the represented state.

The same surface carries the copying obstruction. After zero projectors are removed, trace coordinates and projector synthesis are mutual inverses, so the public record algebra is star-equivalent to complex functions on the active labels. Independently, if one ancilla-free linear isometry of the tensor square copies two sharp states from the same normalized blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so z is zero or one and the overlap-one branch forces equality. Two distinct alternatives copied by one device are therefore orthogonal, which is the finite form of no-cloning on the completed record. The mixed-state no-broadcasting implication is exposed as an explicit adapter premise [6].

At the level of weights the Born form is derived rather than assumed. Every valuation on the effects of the finite surface that is nonnegative, normalized, and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom.

Finite webs do not by themselves supply its hypothesis, and the gap is located exactly. Sharp

two-outcome webs admit the non-Born valuation $F_z(\mathbf{n}) = (1 + n_z^3)/2$; the exhibited unsharp trine excludes it, yet every axis in the finite battery has $n_y = 0$, so the planar response $F_y(\mathbf{n}) = (1+n_y^3)/2$ agrees with the maximally mixed Born weight on the whole battery while staying non-affine off it. The source-attached real S_3 algebraic web is blind to the Pauli-Y direction in the same way: for its noncommuting projector candidates P, Q the commutator phase lift $I/2 - (2\sqrt{3}/3)i(QP - PQ)$ is the $+Y$ projector, and closure under real coarse graining and real Kraus pullbacks stays Y-blind, so a complex phase-sensitive effect is load-bearing for this tomography. The missing step is a source-produced complex-tomographically-complete effect and instrument web with an operational theorem giving full coexistent-effect additivity [6]; outcome frequencies can validate such an instrument and cannot create the universal valuation law.

Boundary 4.3. This is a finite operational record interface. It does not reconstruct an arbitrary quantum state space or an interacting continuum quantum field theory. Its role is narrower: the declared algebra-state representation of the completed consensus record carries the finite probability, conditioning, expectation, and correlation operations consumed by later branches.

5 Overlap defects and edge-center entropy

Overlap agreement has a second layer beyond equality of scalar records. Transport maps can compose only up to a central multiplier. The resulting defect is a finite gluing invariant; removing triangle defects and obtaining endpoint-only transport are distinct operations.

Theorem 5.1 (Central defect strictification and residual holonomy)**.** *Let N_Σ be the nerve of a finite overlap cover. Suppose $U_{ji} = U_{ij}^{-1}$ and*

$$U_{ij}U_{jk} = z_{ijk}U_{ik}, \qquad z_{ijk} \in Z_\Sigma,$$

where the identified coefficient group Z_Σ is abelian and overlap transport acts trivially on it. Then:

(i) z is a Čech 2-cocycle,

$$z_{jkl}z_{ikl}^{-1}z_{ijl}z_{ijk}^{-1} = 1;$$

(ii) its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is invariant under changes of local frame;

(iii) a central edge rephasing removes every triangle multiplier exactly when $[z] = 0$;

(iv) after such a strictification, endpoint-only transport on a charge block $\mathcal{C}_\alpha$ holds exactly when the represented holonomy of every closed loop is the identity on $\mathcal{C}_\alpha$.

Proof. Associativity of $(U_{ij}U_{jk})U_{kl}$ gives the cocycle identity; a local frame change conjugates the multiplier and leaves a central element fixed; an edge rephasing by a 1-cochain changes z by its Čech coboundary; and once the edge maps form a strict 1-cocycle, the discrepancy between two paths with common endpoints is the holonomy of the closed loop they bound. □

A vanishing triangle class does not imply trivial transport around noncontractible loops, and if the center varies as a local coefficient system the cocycle equation must use the twisted Čech differential, outside the untwisted statement above.

5.1 Finite edge-center decomposition

Let a regulated collar be cut into left and right halves with common interface Σ. Suppose finite-dimensional Hilbert spaces $\widetilde{H}_L, \widetilde{H}_R$ carry diagonal unitary actions of a finite compact group G_Σ. Write

$$\widetilde{H}_L = \bigoplus_\alpha V_\alpha \otimes H_{L,\alpha}, \qquad \widetilde{H}_R = \bigoplus_\beta V_\beta^* \otimes H_{R,\beta}.$$

Here the V_α are pairwise inequivalent irreducible unitary representations of G_Σ and $d_\alpha = \dim V_\alpha$.

Theorem 5.2 (Invariant collar decomposition and one-sided edge entropy)**.** *The invariant collar space and its sector-preserving algebra have the exact forms*

$$H_C = (\widetilde{H}_L \otimes \widetilde{H}_R)^{G_\Sigma} \cong \bigoplus_\alpha H_{L,\alpha} \otimes H_{R,\alpha},$$

$$\mathcal{A}_C = \bigoplus_\alpha \mathcal{B}(H_{L,\alpha}) \otimes \mathcal{B}(H_{R,\alpha}), \qquad Z(\mathcal{A}_C) = \bigoplus_\alpha \mathbb{C}\, P_\alpha^C.$$

Let $\Omega_\alpha \in V_\alpha \otimes V_\alpha^*$ be the normalized invariant vector corresponding to $d_\alpha^{-1/2} I_{V_\alpha}$. A sector-preserving collar state has the form

$$\rho_C = \bigoplus_\alpha p_\alpha \rho_\alpha, \qquad \rho_\alpha \in \mathcal{B}(H_{L,\alpha} \otimes H_{R,\alpha}),$$

where $p_\alpha \geq 0$, $\sum_\alpha p_\alpha = 1$, and each block with $p_\alpha > 0$ satisfies $\rho_\alpha \geq 0$ and $\operatorname{Tr} \rho_\alpha = 1$. Embed it in $\widetilde{H}_L \otimes \widetilde{H}_R$ through Ω_α and trace out $\widetilde{H}_R$. If $\rho_{L,\alpha} = \operatorname{Tr}_{H_{R,\alpha}} \rho_\alpha$, the resulting left state is

$$\rho_L = \bigoplus_\alpha p_\alpha \frac{I_{V_\alpha}}{d_\alpha} \otimes \rho_{L,\alpha},$$

and its entropy splits as

$$S(\rho_L) = H(p) + \sum_\alpha p_\alpha S(\rho_{L,\alpha}) + \sum_\alpha p_\alpha \log d_\alpha.$$

The edge term is the expectation in ρ_L of the sector observable

$$Z_L = \sum_\alpha (\log d_\alpha) P_\alpha^L,$$

where P_α^L projects onto $V_\alpha \otimes H_{L,\alpha} \subset \widetilde{H}_L$.

Proof. Schur's lemma on $V_\alpha \otimes V_\beta^*$ gives a one-dimensional invariant space at $\alpha = \beta$ and zero otherwise, which yields the invariant direct sum and the center; the partial trace of $|\Omega_\alpha\rangle\langle\Omega_\alpha|$ over V_α^* is I_{V_α}/d_α; and the entropy of a block-diagonal state is the Shannon entropy of the block weights plus the mean block entropy, each maximally mixed factor contributing $\log d_\alpha$. □

The theorem distinguishes the gauge-invariant collar state from its one-sided reduced state, to which the representation-dimension term belongs. Reading Z_L as an area observable requires a physical realization map, and alignment of the state with the collar decomposition is an additional premise. Related edge-mode and algebra-center decompositions occur in lattice gauge theory [68, 69].

6 Source-derived causal spacetime and the Einstein limit

6.1 How the finite results assemble

The slow-response band supplies an abstract continuous three-dimensional Euclidean carrier under the premises of Theorem 2.3; its unit directions identify with the celestial S^2 and future-null rays, and adjoining a real axis gives the $(1+3)$ Lorentz carrier. Authenticated records separately supply events, generated order and longest-parent-chain height, and the finite placement theorems join the two structures: the edge-speed bound preserves order, and converse support gives order reflection, separation and interval preservation. One common physical refinement family promotes this finite precursor to an effective spacetime when it supplies physical event and link identification, source-selected order/cone embeddings, calibrated count density, independent dimension and manifoldlikeness tests, stable topology, and a unique distinguishing Lorentzian limit; on that same family, convergent tensor-curvature reconstruction or the continuum small-ball/null-balance identification, together with physical stress, Ward/Bianchi conservation, universal coupling and scale calibration, yields the Einstein equation. The populated finite causal precursor is thus an exact result, while the smooth curved $(3+1)$-dimensional manifold and its Einstein dynamics are the stated continuum composition theorem.

6.2 Exact kinematics of the spherical support

The oriented conformal spherical support carries

$$\mathrm{Conf}^+(S^2) \;\cong\; \mathrm{PSL}(2,\mathbb{C}) \;\cong\; \mathrm{SO}^+(3,1), \qquad H^3 \;\cong\; \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 3. \tag{6}$$

These are classical isomorphisms of conformal geometry, consumed as exact mathematics [57]. They identify the three-dimensional hyperbolic homogeneous space of future unit timelike directions. Interpreting those directions as physical observer velocities requires realized events, clocks, and local frames.

Boundary 6.1. Observer-velocity geometry and physical event localization are distinct objects. Authenticated semantic commits supply a finite event carrier and generated order, whose exact longest authenticated-parent-chain height supplies one ordinal coordinate only when events are placed, while a real axis and the source Gram quotient define the ambient carrier with three positive spatial directions. The scaled height coordinate is not an operational clock, the source quotient does not define physical position, and the velocity space H^3 and event localization remain separate constructions.

Theorem 6.2 (Source-derived finite (1+3) causal precursor)**.** *Let L be an authenticated finite semantic-event log with event type E and generated strict order $\prec_L$. Let $h_L : E \to \mathbb{N}$ be its canonical source height: roots have height zero and every nonroot has one plus the maximum height of its authenticated direct parents. Then h_L strictly increases on every parent edge and on $\prec_L$. For every event, it is exactly the maximum length of an authenticated direct-parent chain ending at that event, and a chain attaining the maximum exists. Let V_{src} be the source Gram quotient with positive form g_{src} and real rank three. Define*

$$W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x).$$

Then:

(F1) W_{src} has real dimension $1+3=4$, and Q has one positive and three negative directions;

(F2) every source-unit direction u gives the future-null vector $(1, u)$, and the exact equivalences $\mathsf{SourceUnitDirection} \simeq S^2 \simeq \mathsf{FutureNullRay}$ *identify the boundary of this cone;*

(F3) for any spatial readback $x : E \to V_{\mathrm{src}}$ and scale $\tau > 0$, the placement $F(e) = (\tau h_L(e), x(e))$ sends every authenticated parent edge into the future cone whenever

$$g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \leq \tau^2 \big(h_L(f) - h_L(e)\big)^2$$

on each parent edge $e \to f$; cone transitivity then sends every generated precedence $e \prec_L f$ into the future cone;

(F4) every finite log admits an explicit auxiliary placement of this form: for a finite enumeration $i : E \simeq \mathrm{Fin}(n)$, put the spatial coordinate $i(e)$ along one fixed source axis and take $\tau = n + 1$. This placement is injective, sends reflexive generated ancestry into the future cone, and makes distinct same-height events spacelike; and

(F5) for a placement satisfying the edge-speed condition of (F3), the abstract converse-support condition follows from two concrete finite geometric checks: the spatial readback separates distinct events within each equal-height layer, and whenever $h_L(e) < h_L(f)$ but $e \not\prec_L f$,

$$\tau^2 \big(h_L(f) - h_L(e)\big)^2 < g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)).$$

Consequently every future-causal comparison between placed events is supported by reflexive authenticated ancestry, so $e \preceq_L f$ holds exactly when $F(f) - F(e)$ is future causal. This equivalence and antisymmetry force F to be injective, so it is an exact finite order embedding of the generated poset and preserves every causal interval exactly on its image. Composed with the source-carrier equivalence, it supplies a separated one-chart finite causal-chart interface.

The first two clauses construct the finite (1+3) carrier and signature from the independent real axis and rank-three source carrier; they do not assume a rank-four event chart. Canonical source height appears only in the placements of clauses (F3)–(F4). Clause (F4) removes existence of some separated forward-causal placement as an assumption, while its enumeration is not a source-selected spatial readback. The physical spatial placement, edge-speed condition, converse cone/order equivalence, the two concrete geometric checks, physical interpretation, and clock calibration remain independently checkable data.

Proof. Dimension is additive on the direct sum and positivity of g_{src} gives the inertia; the source-unit/null-ray statement is the exact algebraic equivalence. Induction over authenticated parents bounds every ending chain by source height and constructs an attaining chain. The edge inequality makes each parent displacement future causal, and transitivity of the cone and of ancestry gives the closure statement. For the enumerated placement the enumeration separates events, its diameter is below $n + 1$, and each parent height increment is at least one. For (F5), diagonal pairs follow from reflexivity, decreasing-height pairs are excluded by future orientation, generated pairs are forward causal, and the two geometric checks exclude every remaining unsupported cone comparison; converse support with antisymmetry then forces injectivity, and interval preservation follows at both endpoints. The finite statements are formalized in the geometry proof stack; no limiting topology is used. □

Definition 6.3 (Source-causal continuum certificate)**.** A cofinal family of finite precursors from Theorem 6.2 carries a source-causal continuum certificate when one common family supplies all of the following:

(C1) a physical-event interpretation for the retained commits and an adequacy argument that authenticated support captures the relevant physical signal dependencies;

(C2) source-selected placements satisfying the edge-speed condition and converse support, so that induced cones agree with the generated order in both directions, with link directions dense and isotropic on the source S^2 and ordinal regradings kept separate from operational-clock calibration;

(C3) refinement maps preserving events, generated order, source directions and placements, with a specified notion of convergence;

(C4) a locally calibrated density law $\#I_n/\rho_n \to \mathrm{Vol}(I)$ on causal intervals, with public capacity N excluded unless a theorem identifies it with event number;

(C5) agreement across independent dimension and manifoldlikeness tests, interval-abundance and height/count scaling, and stable thickened-antichain topology or homology, against Poisson-sprinkled $3+1$-dimensional Minkowski and curved controls [35, 36, 37, 40]; and

(C6) convergence of the local cone/overlap data to a distinguishing, time-oriented Lorentzian geometry with the stated uniqueness control for the admitted limit.

This is one certificate with independently falsifiable rows, not a relabeling of a supplied atlas as a derivation.

Theorem 6.4 (Effective manifold and smooth-curvature promotion)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [31, 32, 44, 45]; the local model is (1+3), so a limit passing the dimension and topology rows is an effective four-dimensional time-oriented Lorentzian event manifold. Scalar-curvature convergence alone does not identify the Einstein tensor. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [41, 42], either an explicit discrete tensor-curvature reconstruction converges to the smooth Einstein tensor or the continuum small-ball/null-balance route identifies the limiting finite geometry tensor with G_{ab}, and the stress field and Ward/Bianchi data converge with controlled remainder and calibrated coupling, then the smooth limit is $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$. The tensor identification is not inferred from a finite matrix named geometry.*

The dimensional statements fit together on one carrier: the selected rank-three response quotient supplies the spatial metric, its unit directions are the source S^2, and adjoining a scalar axis gives the $(1+3)$ Lorentz module, with the normalized seam average selected under the naturality and unique-minimizer hypotheses of Proposition 10.5. The following construction realizes a causal order and count measure on source records, and the covariance criterion after it restricts which operational orders can have that geometry.

Theorem 6.5 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let $\varphi = (1+\sqrt{5})/2$, $L = 2/\sqrt{\varphi+2}$ and $q = F_n$, $n \geq 4$. The population*

$$S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0, L]^3$$

consists of q^3 distinct conservative source-record images. At each layer retain this population and let every next-layer event read all preceding-layer sites within $a_q = L/\sqrt{q}$, including itself. Assign

model duration $\Delta_q = a_q/c$, $c > 0$. On bounded model-time windows, the posets have width q^3; each event has edge-count height equal to its layer index. As $q \to \infty$, their order converges away from null separation to the speed-c causal order. With $\rho_q = q^3/(L^3\Delta_q)$, normalized counts converge to $dt\, d^3x$ volume. For fixed interior timelike diamonds,

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q counts strict causally ordered pairs. For vertical duration T, $\mathrm{Vol}(I) = \pi c^3 T^4/24$.

Proof. The exact integer section realizes each site by a conservative seam word. Consecutive Fibonacci approximants permute a uniform reference partition with equal cell volume L^3/q^3 and site-assignment error $H_q = 2\sqrt{3}L/q$; for any h-covering population in a convex window, segment subdivision constructs a k-read path whenever $d \le k(a-2h)$, and every such path satisfies $d \le ka$. Since $H_q/a_q \to 0$ the two cones coalesce, the product cell assignment proves count convergence with null boundaries of zero volume, and the continuum strict-pair integral $\mathrm{Vol}(I)^2/20$ gives the ordering fraction. The path construction, layered order, width and height, the flat constants 1/10, 8/35 and 1/2, the golden cell assignment, the generated interval-count limit and the normalized 1/10 limit are formalized in Lean; the spacetime derivation supplies the source records and explicit volume error bounds [11]. □

Boundary 6.6. This construction supplies a flat effective order and volume for an explicit source-record population, neighbour read law and model clock. At finite regulator the family is produced by the carriers themselves: each carrier holds its record as port loads, reads its position through the rank-three response with the source metric recovered at scale exactly one, decides its neighbours from that readback, and the precedence generated from the carriers' authenticated reads equals the layered order at every level through $q = 89$, that is 704,969 carriers reading one another over ten rounds, with every strict pair counted exactly. On the central diamond the inverted Myrheim–Meyer dimension [38, 39] reads 4.15, 3.61, 4.10, 4.09 and 4.07 at $q = 13, 21, 34, 55, 89$ against 3.0 and 2.0 for the two- and one-dimensional control populations. The ordering fraction reads 0.093 and 0.094 at $q = 55$ and 89 against the flat 1/10; its even-round ladder 0.0600, 0.0803, 0.0887, 0.0932, 0.0955 at $K = 4, 6, 8, 10, 12$ approaches that value with halving increments, the last entry the stratified reading 0.0955 ± 0.0002 of the sampled $q = 144$ level with 2,985,984 sites. The count clock reads 2.05 and 1.96 at $q = 89$ and 144 against the model-time ratio 2. Sampled chain counts and interval-size distributions are retained at $q = 55, 89, 144$, with moving, displaced and lower-dimensional controls. Individual chain-count dimensions near four coexist with nonmonotone interval-distribution discrepancies across these levels. The cumulative distribution comparisons use a finite grid and sampled flat references; joint chain-count covariance and uncertainty for the self-normalized distributions are not supplied by the retained summaries. The $q = 144$ graph-construction driver is not retained, so its pinned summaries do not provide full run reproduction. Off-centre window clipping and unresolved near-cone clock comparisons are retained with the other controls. These finite readings establish neither statistically significant failure of convergence nor a manifold limit. The spacetime derivation gives the individual readings and their uncertainty boundaries [11]. Counts exclude auxiliary read instructions and seam operations. The construction does not select the read law from the observer axioms, calibrate a physical clock, or identify its cone with a matter action. These are distinct requirements of a physical source-causal continuum certificate. With self-waits, the longest history between related endpoints K layers apart has exactly K steps, so uniform duration per step gives moving/rest ratio one rather than $\sqrt{1-\beta^2}$: the ordinal height is not a proper-time instrument. Curved limits, topology stability and the common physical field realization require their stated additional hypotheses.

Theorem 6.7 (Operational selection of the Lorentz cone)**.** *Let E be the source Gram space and $C \subset \mathbb{R} \oplus E$ a nonzero closed convex pointed displacement cone. Suppose the irreducible proper icosahedral rotations preserve C, and boosts along one unit axis preserve it for every rapidity. Then C is the future or past cone of $t^2 - \|x\|^2$. A time orientation selects the future cone.*

Proof. The rotation orbit of the axis spans E. Conjugating its boost and taking finite-matrix product limits gives every boost. The finite rotation average sends (t, x) to $(t, 0)$. A spacelike vector could be boosted to both signs of time, contradicting pointedness. A nonzero causal vector supplies one time-axis ray; boosts and scaling generate its open timelike cone, and closedness supplies the null boundary. Pointedness excludes the opposite orientation. The spacetime derivation gives the product-limit proof [11]. □

Here covariance must preserve possible influence on actual histories, interventions and clock readouts. An algebraic Lorentz action on source directions alone supplies no such operational transformation. The criterion selects cone shape under that hypothesis; it does not select read weights or a temporal field equation.

Corollary 6.8 (Proper-time ratios from record counts)**.** *In Theorem 6.5, let I_q, J_q approach interior timelike diamonds of proper durations $\tau_I, \tau_J > 0$. Then the observer readout*

$$R_q(I; J) = \left(\frac{|I_q|}{|J_q|}\right)^{1/4} \longrightarrow \frac{\tau_I}{\tau_J}.$$

Within the class of positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.

Proof. Diamond volume is $\pi c^3 \tau^4/24$; the common count weight and constant cancel in the ratio. For uniqueness, $g(t) = F(\pi c^3 t^4/24)$ is positive and additive, hence increasing and linear by rational approximation. □

The readout uses authenticated ancestry and counts, with a reference interval setting its unit. It reconstructs elapsed duration after the upper endpoint, supplies a geometric clock for the declared source family independently of a Hamiltonian or oscillator, and identifying a running matter clock with it requires their common physical realization.

What the read law fixes and what it leaves free. The declared read law is sufficient for the limit and not necessary [11]. Explicit sparse stencils with at most $24n^2 + 6n + 9$ directions at $q = 4^n$ give the same flat causal order, count volume $\pi c^3 T^4/24$, ordering fraction $1/10$ and count clock, while their crossing-read entropy across a fixed plane section falls to zero relative to the complete-neighbour law, and a coordinate-plus-sparse family makes the per-area entropies of two sections differ by a factor $\sqrt{2}$ in the limit; the causal and count limits therefore select no crossing-entropy law. A second sparse family with $\Theta(q^{9/8})$ parents per event preserves it exactly: every regular plane section Σ has crossing entropy $\pi q^4 |\Sigma|/(4L^2)$ to leading order, one nat per unordered crossing pair, and thinning at the original radius cannot retain those counts. The crossing functional has a variational limit only with a short-range repair. The plain residue partition has vanishing entropy and the sub-isoperimetric limit $2^{-1/3}(36\pi)^{1/3} v^{2/3}$; adding reads inside a ball of radius r with $r^4/(mq) \to \infty$ and $r^4/q^2 \to 0$ restores compactness of bounded-entropy families and gives the Γ-limit to Euclidean perimeter, and when a ball of volume v fits strictly inside the domain the minimum values tend to $(36\pi)^{1/3} v^{2/3}$ with every limiting minimizer a ball. On the quantum side, a degenerate two-state band of positively timed register flights has a matrix of first-order Pauli

velocity coefficients of nuclear norm at most c; isotropic coherent transport therefore has speed at most $c/3$, attained by the minimal tetrahedral process. Its explicit massive extension keeps the full eight-channel spectrum and, for $0 < m\tau < \pi/2$, has an all-band speed bound $(c/3)\cos(m\tau)$, where τ is the flight duration. At fixed resolution and duration, sampled compact H^2 CAR number reads and phase encodings outside the speed-$c/3$ Dirac cone have influence $O(\sqrt{a})$ uniformly over states and spectator ancillas, although an exact cell-resolving one-flight record retains speed c. Two coherently labelled mass packets give a positive local interference readout with reference beat $\Delta m/\gamma$ in native flight time, $\gamma = (1 - |u|^2/(c/3)^2)^{-1/2}$, with bounded finite preparation, evolution and detector errors. A finite analytic witness retains click-probability swing above 0.6753 through a full moving cycle; its much finer spacing is not an executed simulation. A separate declared coherent-code source constructs the native gates, preparations and full fermionic operations needed for this clock. Coherent code gluing is an additional transport premise: a classical completion of the same classical-record grammar does not imply it. The masses, physical source identification and clock calibration remain supplied [11].

Quantum operations with live noisy records. The source also supports a joint noisy implementation with faulty runtime feedback and a maintained public archive. Complete transfer/reset channels reduce full six-dimensional carrier leakage locally before error correction; private Kraus branches, public outcomes and aborts are retained. For fixed nonzero subthreshold elementary error strength, a powered expander refreshes $O(\log q)$-bit words and staged decoding exports from $O(\log\log q)$ protection levels. In the empty-vacuum experiment the joint error is $O(q^{-20}\log^{16+2\kappa}(2q))$ and diagnostic storage-time costs $O(q^6\log^{25+3\kappa}(2q))$, with fixed library constant κ. These are conditional bounds using imported fault-tolerance and expander theorems and a joint local fault-norm hypothesis. Finite executable controls check interfaces, decoding and faults; the arbitrary-size archive proof is analytical, with large constants and no measured hardware threshold.

In a separate filled-vacuum experiment, the same reflecting massive walk has spectral gaps at $+1$ and -1. Filling the negative modes of its principal signed logarithm h gives the positive excitation generator $H_{\mathrm{sea}} = \mathrm{d}\Gamma(h) - \mathrm{Tr}(hP)I$, where $P = \mathbf{1}_{h<0}$. It generates exactly the original integer-time Fock channels. Local covariances converge with both carrier bands retained. For the stated smooth carrier kernels, local operations and complete number reads have uniformly bounded mean excitation cost. A translated-mode read retains the occupied background and has a finite analytical probability-error bound below 0.038 and swing above 0.923; the corresponding enormous apparatus is not simulated. The longer deterministic sea preparation raises storage-time to $O(q^{27}\log^{55+3\kappa}(2q))$, with joint protected error $O(q^{-55}\log^{36+2\kappa}(2q))$. An explicit high-band Floquet shift preserves the vacuum, integer-time channels and infrared energy limits while changing ultraviolet costs. Source admission, external scheduling, growing controls and physical energy calibration remain hypotheses; neither construction supplies autonomous time or drive work [11].

7 Einstein reconstruction

Even a complete calibrated scalar-clock response does not determine light bending. On a positive local two-field action, the family $E = [u^{\mathsf{T}}Ku + \kappa(v-\gamma u)^{\mathsf{T}}K(v-\gamma u)]/2 - j^{\mathsf{T}}u$, with $K > 0$, $\kappa > 0$ and arbitrary real γ, has identical clock response and clock marginal, while its stationary spatial field is $v = \gamma u$. One shared metric and matter/Maxwell coupling gives identical redshift and exterior bending $2(1+\gamma)GM/(c^2 b)$ to first order in the weak field at fixed finite γ. The conformal member $\gamma = -1$ has exactly zero bending. The linearized physical exterior tensor equation selects $\gamma = 1$;

scalar vacuum and Ward balance do not. Constitutive fields, sources and probe identification are declared, and this comparison supplies no physical Einstein-branch entry [11].

Theorem 7.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η. Suppose a symmetric form X satisfies $X(k,k)=0$ for every null vector k. Then $X=\phi\eta$ for a scalar ϕ. Null-null values therefore determine a symmetric tensor modulo the metric line. In an η-orthonormal basis with $\eta=\mathrm{diag}(-1,1,1,1)$, set $s=\sqrt{3}/3$ and take the nine null vectors*

$$\begin{gathered}(1,\pm1,0,0),\quad (1,0,\pm1,0),\quad (1,0,0,\pm1),\\ (1,s,s,s),\quad (1,s,s,-s),\quad (1,s,-s,s).\end{gathered}$$

On the η-trace-free representative, use coordinates

$$x=(X_{00},X_{01},X_{02},X_{03},X_{11},X_{12},X_{13},X_{22},X_{23}),\qquad X_{33}=X_{00}-X_{11}-X_{22}.$$

The corresponding nine-charge design reconstructs the quotient

$$\mathrm{Sym}^2(V^*)/\mathbb{R}\eta.$$

Its determinant is $8192/27$*. With the supremum norms on coordinates and charges, its exact decoder obeys*

$$\|x\|_\infty\le(2+\sqrt{3})\|q\|_\infty.$$

Proof. With $\eta=\mathrm{diag}(-1,1,1,1)$ and future null rays $(1,n)$, $|n|=1$, the identity $X_{00}+2X_{0i}n_i+X_{ij}n_in_j=0$ holds on the unit sphere; its odd part gives $X_{0i}=0$, its degree-two harmonic part gives $X_{ij}=\phi\delta_{ij}$, and the constant part gives $X_{00}=-\phi$, so $X=\phi\eta$ and the quotient has dimension nine. Substituting the nine displayed directions into the trace-free coordinate basis gives determinant $8192/27$, and the explicit decoder has maximum absolute row sum $2+\sqrt{3}$. The Lean development proves the nine directions null and formalizes the design map, an explicit left-inverse decoder, injectivity, and the metric-line ambiguity [1, 11]. □

Nine readings interpolate a tensor representative; they do not establish that an arbitrary directional family is tensorial or represents physical stress. Additional directions test the dependent-family relations within a declared measurement and numerical error budget. For a general design, resolved rank nine is required. With rays normalized to $k^0=1$ and a Frobenius-orthonormal trace-free basis, charge error δ contributes at most $\delta/s_{\min}$ in tensor Frobenius norm, where $s_{\min}$ is the design's smallest singular value; geometry and solver errors are separate. No finite sample alone controls unsampled directions.

The four-dimensional small-ball coefficients of entanglement-equilibrium arguments [64] are isolated from their physical reading by one machine-checked identity: with $G_N>0$ and $\ell>0$, scalars satisfying

$$\delta S_{\mathrm{bulk}}=\frac{8\pi^2\ell^4}{15}\,t,\qquad \delta A=-\frac{4\pi\ell^4}{15}\,f,\qquad \delta S_{\mathrm{bulk}}+\frac{\delta A}{4G_N}=0 \tag{7}$$

obey $f=8\pi G_N t$ exactly, substitution cancelling the common nonzero prefactor.

The tensor completion begins at an explicitly realized null-balance relation. This is the point at which the physical bridge enters the mathematical implication.

Theorem 7.2 (Tensor completion of a realized null balance)**.** *Let (M,g) be a connected, time-oriented, C^3 Lorentzian four-manifold with Einstein tensor $G^{(g)}_{ab}$. Let T_{ab} be a symmetric C^1 tensor*

satisfying the independent Ward premise $\nabla^a T_{ab} = 0$. *Suppose a common realized tower supplies a constant* $\kappa > 0$ *such that, at every point and for every null vector* k,

$$G^{(g)}_{ab} k^a k^b = \kappa T_{ab} k^a k^b, \tag{8}$$

and suppose $G_N > 0$ *with* $\kappa = 8\pi G_N$. *Then there is a constant* Λ *such that*

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab} \tag{9}$$

on M. *If a reference event* p_0 *and scalar* Λ_0 *satisfy*

$$(G^{(g)}_{ab} - 8\pi G_N T_{ab})\big|_{p_0} = -\Lambda_0 g_{ab}\big|_{p_0},$$

then $\Lambda = \Lambda_0$.

Proof. At each point, Theorem 7.1 applied to $G^{(g)} - \kappa T$ gives a scalar ϕ with $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity $\nabla^a G^{(g)}_{ab} = 0$, the Ward premise, and metric compatibility give $\nabla_b \phi = 0$. Connectedness makes ϕ constant. Set $\Lambda = -\phi$ and use $\kappa = 8\pi G_N$. Evaluation at p_0 gives $\Lambda = \Lambda_0$ under the final calibration premise. □

For OPH to supply the null-balance premise (8), one common refinement tower must realize all of the following:

(G1) the effective manifold of Theorem 6.4, with sufficient regularity for its Levi–Civita connection and contracted Bianchi identity;

(G2) geometrically normalized cap modular flow and half-sided modular inclusions whose directional charges define one symmetric tensor T_{ab};

(G3) the Ward identity for that tensor, independently of its reconstruction from directional charges;

(G4) the edge-center split of Theorem 5.2, the physical normalization $\delta\langle Z_L\rangle = \delta A/(4G_N)$, the modular first law, and generalized-entropy stationarity;

(G5) one family with $\ell_r > 0$, $\ell_r \to 0$, all named remainders $o(\ell_r^4)$, the continuum diamond-kernel and fixed-volume area formulas used in the small-ball identity (7), and enough local observer directions to establish (8); and

(G6) universal coupling of the stress and entropy branches, a vacuum reference, and a physical scale calibration establishing $\kappa = 8\pi G_N$.

The exact finite layer contains null tomography, the edge-entropy split, and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise while separately checking the small-ball arithmetic; the bridge between them is (M4). One Newtonian ingredient is a machine-checked shell law on the committed carrier: a spherically symmetric flux with constant shell charge falls off exactly as the inverse square, the exponent being the carrier dimension minus one with the dimension three supplied by Theorem 2.3. Its flux premises are declared, and its join to the composed Einstein branch is unbuilt.

8 Two gauge reconstructions and a finite matter image

8.1 Sector reconstruction

The overlap and edge-center branch gives a structural gauge result before a particular compact group is identified.

Theorem 8.1 (Bosonic sector reconstruction)**.** *Suppose a cofinal refinement tail carries zero-obstruction, trivial-holonomy bosonic edge sectors with compatible fully faithful pullback functors, that their closure* Sect_∞ *is an essentially small, additive, idempotent-complete, semisimple rigid symmetric* C^**-tensor category with simple unit and finite-dimensional Hom spaces, and that it carries a faithful, unitary, strong symmetric monoidal,* $*$*-preserving fiber functor* $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$*. Then* $G_{\mathrm{Tan}} = \mathrm{Aut}^{u,*}_{\otimes}(\mathcal{F})$ *is compact in the topology of pointwise operator convergence,* $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G_{\mathrm{Tan}})$ *as symmetric* C^**-tensor categories, and the reconstructed group is unique up to isomorphism for the specified category and fiber functor.*

Proof. Each unitary monoidal natural $*$-automorphism of $\mathcal{F}$ has a component in the compact group $U(\mathcal{F}(X))$ at every object, and naturality, tensor compatibility, symmetry and $*$-compatibility are closed equations, so the automorphism group is a closed subgroup of $\prod_X U(\mathcal{F}(X))$ over a small skeleton and is compact. Doplicher–Roberts/Tannaka reconstruction gives the equivalence [71, 72], and a second compact group compatible with the same fiber functor is identified with the same automorphism group. □

The theorem classifies the group encoded by the realized sector data. A trivial sector category reconstructs a trivial group, so the statement selects the Standard Model group only given a suitable sector witness, and it identifies its group with the finite response group of the next subsection through a separate premise.

8.2 The twelve-port response algebra

Let A be the icosahedral adjacency operator and J the antipodal involution on the twelve ports. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I. \tag{10}$$

Put $\varphi = (1+\sqrt{5})/2$. Choose from each antipodal axis one of the certificate's exact coordinate vectors $u_i \in \mathbb{Q}(\sqrt{5})^3$, each with $\|u_i\|^2 = 2+\varphi$, and put $U = [u_1 \ \cdots \ u_6]$. For a real port field f, write

$$b_i = \frac{f_i + f_{Ji}}{2}, \qquad d_i = \frac{f_i - f_{Ji}}{2}, \qquad c = \frac{1}{6}\sum_i b_i, \qquad b^0 = b - c\mathbf{1}.$$

The even coordinates split as $\mathbf{1} \oplus \mathbf{5}$. The odd coordinates split as the rank-three frame channel detected by U and its Galois-conjugate companion detected by $\sigma(U)$, where $\sigma(\sqrt{5}) = -\sqrt{5}$, giving the source-module decomposition

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'.$$

Theorem 8.2 (A1–A2 gauge Lie-type forcing)**.** *Let* $D : P_{12} \to \mathfrak{u}(H)$ *be the complete reversible response derivative supplied by A1, and put* $\mathfrak{g} = D(P_{12})$*. Suppose the endogenous transport clause of A2 covers every proper carrier action* $a \in A_5$ *and gives projective implementers satisfying*

$$\mathrm{Ad}_{[U_a]}\, D(v) = D(a\cdot v), \qquad [U_a] = [g_a c_a],$$

where $g_a \in G_D^0$ and c_a centralizes $\mathfrak{g}$ pointwise. Then

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The center is the unique trivial line. Up to exchanging **3** *and* $\mathbf{3}'$ *by the outer automorphism of A_5, the $\mathfrak{su}(2)$ ideal carries* **3** *and the $\mathfrak{su}(3)$ ideal carries* $\mathbf{3}' \oplus \mathbf{5}$.

Proof. Injectivity of D gives $\dim \mathfrak{g} = 12$. Its commutator-closed image is a finite-dimensional Lie subalgebra of $\mathfrak{u}(H)$, and the positive trace pairing makes it compact. For every $a \in A_5$, the centralizer factor c_a acts trivially on $\mathfrak{g}$, so $\mathrm{Ad}_{[U_a]}|_{\mathfrak{g}} = \mathrm{Ad}_{g_a}|_{\mathfrak{g}}$ is inner. The chosen projective lifts need not form a homomorphic section. Covariance through the injective map D nevertheless gives the homomorphism

$$a \longmapsto (D(v) \mapsto D(a \cdot v)) \quad \text{from } A_5 \text{ to } \mathrm{Int}(\mathfrak{g}).$$

The transitive proper-carrier action fixes only the uniform port line, hence $\dim \mathfrak{g}^{A_5} = 1$.

Compactness gives the reductive decomposition $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$. Inner automorphisms fix the center pointwise, so $\dim \mathfrak{z} \leq 1$.

If $\mathfrak{z} = 0$, the low-dimensional classification of compact simple Lie algebras forces a twelve-dimensional compact semisimple algebra to be $\mathfrak{su}(2)^4$ [70]. Inner automorphisms preserve each simple ideal. On each three-dimensional ideal, the restriction of the A_5 action is either trivial or faithful because A_5 is simple. Its fixed-space dimension is therefore three or zero: a faithful A_5 subgroup of SO(3) cannot fix an axis, since a finite rotation group fixing an axis is cyclic. The fixed-space dimension of $\mathfrak{g}$ would be a multiple of three, contradicting the one-dimensional fixed line. Hence $\dim \mathfrak{z} = 1$.

The semisimple part has dimension eleven. The same low-dimensional classification leaves only $11 = 8 + 3$, realized by $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$. Adding the center proves the Lie-algebra claim. The center must be the unique fixed line. The three-dimensional simple ideal cannot carry the trivial action, since that would add three fixed directions; hence it carries **3** or $\mathbf{3}'$, and the eight-dimensional ideal carries the complementary triplet together with **5**. □

Boundary 8.3. No ambient compact group or candidate Lie algebra occurs among the premises of Theorem 8.2: A1 supplies the faithful compact commutator-closed response algebra, A2 supplies its inner proper-carrier action, and the theorem fixes the abstract local Lie-algebra type. The port count twelve equals the dimension of the forced algebra, so A1 fixes that dimension by declaration; the content of the theorem is the exclusion, through compactness and the one-dimensional fixed space, of the only other twelve-dimensional compact type $\mathfrak{su}(2)^4$, together with the forcing of the unique center. Identifying antipodal ports instead gives a six-port projective-plane carrier: the same compactness, inner A_5 action and single-fixed-line requirements admit no six-dimensional current algebra. This changes both topology and port count and does not isolate orientation as the cause. The source receipts derive the carrier module, the inverse-port response and its relative band signs, while the reconstruction of D from ordered reversible histories or same-current holonomy is consumed as a premise, and the branch supplies no physical matrix current, coupling, matter action, global quotient, or laboratory attachment (M3).

Two exact negative controls bound how a source realization could be built: for the declared adjacency recurrence and inverse-port readback the full response-word algebra is the commutative span of I, A, A^2, A^3, with neither twelve independent generators nor a nonzero bracket, and the complete target-free alternating-bracket search space $\mathrm{Hom}_{A_5}(\Lambda^2 \mathbb{Q}^{12}, \mathbb{Q}^{12})$ has dimension fourteen, with its real compact locus classified into P, F, G conditional on three named compact-Lie inputs [13, 2, 1]. No uniqueness claim over carriers is made, and the closure reading of the carrier enters no theorem here.

Let

$$\Phi_0(b^0) = \sum_{i=1}^{6} b_i^0 u_i u_i^{\mathsf{T}}, \qquad \widehat{x}\, y = x \times y.$$

The conditional matrix witness uses the space

$$H = \mathbb{C}^3_E \oplus \mathbb{C}^3_W$$

and four nonzero rational coefficients $\lambda_1, \lambda_5, \lambda_3, \lambda_{3'}$, one on each source band. These are explicit branch premises. The two three-dimensional response blocks match the two rank-three odd channels; they are not asserted to be a derived physical Hilbert space. Write v_p for the signed coordinate vector at port p. For $g \in A_5$, let $R_g \in \mathrm{SO}(3)$ be the exact rotation satisfying $R_g v_p = v_{g(p)}$ on the oriented port frame, and define

$$\Pi(g) = \mathrm{diag}(R_g, \sigma(R_g)).$$

The target A_5 action on $\mathfrak{u}(H)$ is conjugation by $\Pi(g)$.

Theorem 8.4 (Conditional four-band finite port-response algebra)**.** *On the declared twelve-port carrier and supplied response representation, define*

$$\begin{aligned} K(f) = \mathrm{diag}\big(\lambda_3 \widehat{Ud} + i\{\lambda_1 c I_3 + \lambda_5 \Phi_0(b^0)\}, \\ \lambda_{3'} \widehat{\sigma(U)d}\big) \in \mathfrak{u}(H). \end{aligned} \tag{11}$$

Then K is injective, its image is commutator-closed, and

$$K(g{\cdot}f) = \Pi(g) K(f) \Pi(g)^*.$$

The induced A_5 action on $\mathrm{im}\, K$ *is inner. Let $K^{-1} : \mathrm{im}\, K \to P_{12}$ denote the inverse of K onto its image. With the bracket pulled back from that image,*

$$[x, y]_K = K^{-1}([K(x), K(y)]_{\mathrm{mat}}),$$

one has

$$(P_{12}, [\,\cdot\,,\cdot\,]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2). \tag{12}$$

Its center is the uniform one-dimensional port line, its derived algebra has dimension eleven, and the five-dimensional A_5 band is noncentral. The map has a positive-definite invariant Hilbert–Schmidt pullback.

Proof. Evaluating (10) on the four adjacency eigenspaces gives the antipode relation, and the source protocol solves the common farthest-shell filter to give the signed response $-J$. The constant coordinate maps to $i\mathbb{R}I_3$; the six rank-one axis matrices span the symmetric 3×3 matrices with $\sum_i u_i u_i^{\mathsf{T}} = (5{+}\sqrt{5})I_3$, so their sum-zero coefficients map isomorphically onto the traceless symmetric matrices under Φ_0; U is an isomorphism on the **3** band and annihilates **3′**, with $\sigma(U)$ complementary. Hence K has rank 12 and image exactly $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Equivariance, positive definiteness of the invariant pairing, and the statement that $\Pi(g)$ is the exponential of an element of $\mathrm{im}\, K$ are exact calculations in $\mathbb{Q}(\sqrt{5})$ over the full sixty-element domain, part of the proof by the exhaustive-computation rule of Table 1 [1]. □

Boundary 8.5. The A_5-module decomposition alone does not determine a Lie bracket: the same module also carries the zero bracket, whose trivial response-generated group fails the A2 endogeneity clause. The independent inputs to Theorem 8.4 are the carrier incidence and orientation, the

supplied response representation, and four nonzero band coefficients. The block architecture and coefficients are declared. No measured particle, mass, or coupling value enters the exact verification. The theorem constructs the pulled-back bracket for this response map; it does not select K from source histories or prove a uniqueness claim over all A_5-equivariant brackets or finite carriers. The abstract Lie type is forced independently by Theorem 8.2.

8.3 Global form and anomaly-free exterior module

Theorem 8.6 (Finite matter image and exhaustive exterior-menu selection)**.** *Let*

$$V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$$

be an additional, explicitly supplied trace-balanced matter representation of the Lie algebra forced in Theorem 8.2. This $3 \oplus 2$ *matter space is distinct from the supplied response space* $H = \mathbb{C}^3_E \oplus \mathbb{C}^3_W$ *and is not derived from it. Let*

$$\widetilde{G} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$$

act on V *by*

$$(g_3, g_2, z) \cdot (v_3, v_2) = (z^{-2} g_3 v_3, z^3 g_2 v_2).$$

The differential of this action gives the displayed hypercharges. Its exterior module has the exact branching

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L \tag{13}$$

with

component	$\mathrm{SU}(3) \times \mathrm{SU}(2)$ *type*	Y
Q	$(3, 2)$	$1/6$
u^c	$(\bar{3}, 1)$	$-2/3$
e^c	$(1, 1)$	1
d^c	$(\bar{3}, 1)$	$1/3$
L	$(1, 2)$	$-1/2$

This is the familiar one-generation $\mathbf{10} \oplus \overline{\mathbf{5}}$ *branching associated with* $\mathrm{SU}(5)$ *unification [73]. It has total complex dimension fifteen. All perturbative gauge and mixed gravitational anomalies vanish, and the number of weak doublets, counted with color multiplicity, is four. The common kernel of the action of* $\widetilde{G}$ *on* V*, and hence on* $\Lambda^2 V \oplus \Lambda^4 V$*, is* $\mathbb{Z}_6$*. Hence the maximal faithful matter image is*

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{14}$$

Define

$$G_{\mathrm{packet}} := S(U(3) \times U(2)),$$

the maximal faithful image of the displayed declared exterior module. This quotient is familiar as a possible global form of the Standard Model gauge group [75]. The scan menu consists of the ten nontrivial irreducible summands of $\Lambda^\bullet V$ *after the invariant vacuum* $\Lambda^0 V$ *and invariant top line* $\Lambda^5 V$ *have been excluded by declaration. Among all* $2^{10} = 1{,}024$ *subsets of this menu, imposing nonemptiness, chirality, and vanishing of the mixed gravitational–*$\mathrm{U}(1)$*,* $\mathrm{SU}(3)^2\mathrm{U}(1)$*,* $\mathrm{SU}(2)^2\mathrm{U}(1)$*, and* $\mathrm{U}(1)^3$ *traces leaves exactly two rank-fifteen masks. They are exchanged by charge conjugation, and the exterior-parity grading is computed as an output.*

Proof. The exterior-sum identity $\Lambda^k(A \oplus B) = \bigoplus_{p+q=k} \Lambda^p A \otimes \Lambda^q B$ gives the displayed five components and charges, and the anomaly-cancellation conditions are standard [74]. With the conventional quadratic indices $T(3) = T(2) = 1/2$ the pure color anomaly is $2 - 1 - 1 = 0$, the two color fundamentals sitting in the weak doublet against the antifundamentals u^c and d^c; the local $\mathrm{SU}(2)^3$ anomaly vanishes identically because the doublet is pseudoreal; and the mixed sums are $\frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0$ and $\frac{1}{4} - \frac{1}{4} = 0$. The abelian cubic and mixed gravitational sums are

$$6\left(\tfrac{1}{6}\right)^3 + 3\left(-\tfrac{2}{3}\right)^3 + 1 + 3\left(\tfrac{1}{3}\right)^3 + 2\left(-\tfrac{1}{2}\right)^3 = 0, \qquad 6\left(\tfrac{1}{6}\right) + 3\left(-\tfrac{2}{3}\right) + 1 + 3\left(\tfrac{1}{3}\right) + 2\left(-\tfrac{1}{2}\right) = 0,$$

and with three color copies of Q against one L the Witten parity is even [76]. For the displayed cover action the element $(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3})$ acts trivially on every row and generates a cyclic group of order six; direct center-action enumeration excludes a larger common kernel, and the independent integer presentation has Smith invariants $(1, 1, 1, 1, 1, 6)$. The cover action has determinant one on V, so $\Lambda^4 V \cong V^*$ equivariantly and its kernel on the exterior module equals its kernel on V, which gives (14). The finite scan evaluates nonemptiness, chirality, and the four displayed traces on all 1,024 masks of the declared menu; its only survivors are the two exterior-parity masks, exchanged by the explicit conjugation permutation, with parity recorded after selection rather than used as a selection predicate. □

The structure layer composes as one conditional statement: a single theorem on the declared premise bundle proves the forced Lie type, the unique type-matched nearest family at the bundle's balanced point, and the fifteen-state count, anomaly forms and $\mathbb{Z}_6$ stabilizer directly on the bundled selection mask, with exactly two masks surviving at the selected coarse G label and the mask permutation exchanging them. It selects no global form, physical matter carrier, current or action, so it is not a two-model uniqueness theorem for complete Standard Model structures. One corollary concerns a channel the framework does not contain: for the forced type, instantiated by the declared charged-double-triplet current witness, the adjoint is $(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0)$ and contains no mixed $(3, 2, -5/6)$ generator or conjugate, so the minimal simple grand-unified gauge-exchange channel is absent from the twelve-dimensional current algebra. This algebraic statement establishes no proton lifetime and excludes neither scalar mediators nor higher-dimensional baryon-number violation.

Boundary 8.7. The finite response group G_{packet} and the sector-reconstructed group G_{Tan} arise from different data, and their equality is a separate commuting-square premise. Exterior parity does not select physical exchange statistics. The theorem classifies a finite declared exterior representation and its maximal faithful image; laboratory current attachment, physical matter identification, exclusion of extra light sectors, scalar attachment, family multiplicity, Yukawa couplings, mixing, masses, and a continuum quantum-field realization are unsupplied, and the exclusion of $\Lambda^0 V$, $\Lambda^5 V$, direct sums, vectorlike additions, and neutral singlets is underived by this scan. Section 9.7 states term by term what the reconstruction supplies for the textbook Lagrangian and what it consumes.

8.4 Massless carriers, conditionally

Three conditional masslessness statements occur on distinct premise sets, each a consistency consequence of its declared branch and none an independent discriminator from the Standard Model or General Relativity. For the photon, a finite one-loop calculation importing the electroweak action and its symmetry-restored vacuum relation certifies a transverse self-energy and a null direction of the neutral mass matrix, so that fixture carries no hard photon mass term; a derivation additionally requires a source-generated action, (M3), and a photon-pole map. For the gluon, the color

factor in (12) lies in the unbroken subalgebra of the conditional global form, and under the separately declared scalar completion no scalar direction carries color, so the finite action contains no hard color-vector mass term, with the physical gluon spectrum and confinement outside the finite calculation. For the graviton, linearizing the composed relation of Theorem 7.2 about a suitable vacuum yields a null spin-two kernel with two transverse-traceless modes, conditional on the same premise vector and supplying no source-selected Hilbert space, positive-residue pole, or rest-mass prediction.

8.5 The Yang–Mills mass gap

Confinement sits on the other side of the same gauge sector, and the finite repair dynamics reaches it as a conditional identity rather than as a spectrum calculation. The Clay problem asks for a nontrivial four-dimensional quantum gauge theory with a positive mass gap for every compact simple gauge group [19]. The companion paper states what the architecture supplies and what it consumes [18].

The finite mechanism is exact. On the support-visible compact-gauge branch, local holonomy data and the screen scaling chart give the candidate Euclidean form, and exact local repair acts on it as a positive Euclidean relaxation generator at finite cutoff. Atomic heat-bath collars supply that generator, and a finite source-type table together with a uniform approximate-tensorization receipt gives the explicit rate floor $\delta_\star = c_\star/A_\star$.

The Lean statement is narrower: commuting collar projections and strictly positive rates are hypotheses, the scalar relaxation lemma is checked separately on a homogeneous fiber, and the minimal fixed-space witness instantiates abstract finite operator premises only. It is not a compact-gauge construction, a continuum certificate, or a physical mass-gap derivation.

Boundary 8.8. The four-dimensional conclusion consumes convergence of the Schwinger functions, reflection positivity, Euclidean locality, nontriviality, and convergence of the transfer data. Under those certificates the continuum Yang–Mills Hamiltonian gap is identified with the limiting repair gap, so the statement is a conditional implication whose hard content sits in the continuum construction. The finite repair theorem is separate from the Clay claim and carries no continuum limit.

A finite $\mathbb{Z}_2$ diagnostic at $L = 2, 3$ finds the Wilson constant-rate receipt failing on the tested interacting grid while the Kogut–Susskind Doob transform has exact positive fiber-dependent rates $c_\ell = \lambda(r + 1/r) \geq 2\lambda$; quotient-space approximate tensorization and continuum transfer are unconstructed.

9 Position, energy, matter, and the action

A reconstruction that starts from records owes an account of the objects physics starts from. This section states what position, energy, matter, and the action are inside the finite architecture, which of them are derived and which are declared, and how far the route to a Standard Model Lagrangian is carried. The layer boundary is sharp throughout: the finite objects are constructed, and their identification with laboratory position, laboratory energy, physical matter, and a physical action consumes the maps of Section 1.3.

9.1 Position

Position is the carrier's own readback of its accumulated records. Theorem 2.3 makes this exact: the normalized repair response has intrinsic rank three, the signed record module and the seam current

module embed densely in one abstract three-dimensional Euclidean carrier under the response-selected Gram metric, cumulative records act on it by exact internal translations, and the sixty proper carrier maps act by isometries. Displacement is a record difference, and the dimension of the carrier is an output of the incidence rather than an input; physical position, physical length, and cofinal gluing between carriers are unsupplied.

9.2 Energy

Energy is the modular ledger of the record state. For a reference τ the modular Hamiltonian is $K = -\log \tau$, the second law of Section 11 appears as $\Delta S \geq \Delta\langle K\rangle$ on τ-preserving channels, and the exact identity

$$S(p) - S(\tau) = \langle K\rangle_p - \langle K\rangle_\tau - D(p\|\tau)$$

makes the entanglement first law $\delta S = \delta\langle K\rangle$ a first-order corollary. The quantity $\langle K\rangle$ is the internal energy ledger every later statement weighs.

That ledger composes exactly. Against a positive product reference it is additive over parts for every joint state, correlated or otherwise, and the binding defect against a general joint reference is the expectation of $\log(\tau_1\tau_2/\tau_{12})$, which vanishes on the product reference. Against the Gibbs reference of a declared total $H_1 + H_2 + V$ with part references at H_1 and H_2 and one inverse temperature, the defect is $\beta\langle V\rangle + \log(Z_{12}/Z_1Z_2)$, whose constant vanishes at $V = 0$. For a declared attractive $V \leq 0$ at $\beta \geq 0$ the state part is nonpositive and the constant is nonnegative, so attraction alone leaves the sign of the defect unfixed; the two-point uniform state has the exact positive defect $-1/4 + \log((3 + e)/4)$.

The mass–energy identity has two halves here. The kinematic half is a normalization: for a declared positive mass parameter along a future unit timelike frame direction of the Hermitian Lorentz module, the scalar coordinate satisfies $E^2 = m^2 + |p|^2$ and equals m exactly at vanishing spatial coordinates, every oriented Lorentz map transports the construction with the same Lorentz square, a future null vector satisfies the zero-mass identity, and the declared plane wave of that momentum advances in phase by exactly m per unit proper time along its own worldline. The dynamical half is the slope with which $\langle K\rangle$ enters inertia, and two independently declared selection routes constrain it. On the momentum map, in the family $P_\lambda = (m + \lambda E)\, u + (1 - \lambda)E\, e_0$ every member has rest energy $m + E$, yet P_λ is Lorentz covariant and on the shell of its rest energy in every frame exactly when $\lambda = 1$ or $E = 0$, the converse witnessed by an explicit boost. On the action, midpoint refinement of a worldline leaves the proper length invariant and doubles the step count, so for $E \neq 0$ refinement invariance of $E(aL + bM)$ forces $b = 0$ and excludes the per-step additive ledger, and the declared proper-time principle selects coefficient one in the family $E\lambda L$; the length action $(m + E)L$ then has equation of motion $(m + E)\,\Delta T =$ impulse on timelike windows, and the inertial coefficient of a composite is $m_1 + m_2 + E_1 + E_2$ plus the binding defect, the shape a mass defect takes.

Once the seam-step worldline unit is identified with the scaled field step, the stability window of Section 9.5 carries a dimensionless causal bound: inside the window every per-step crossing is spacelike, a block of k rests and one crossing is timelike exactly when $2 < (k + 1)h$, which forces at least two rests per crossing, and timelike block speeds approach one from below without attaining it. The identification is declared; without it the two thresholds are independent and no signal speed is claimed.

Boundary 9.1. The slope family, the covariance requirement, the refinement, and the proper-time principle are declared, and every shape is a declared enrichment under the non-identifiability theorem of Section 9.4. Nothing here derives a mass from repair data. Reading $\langle K\rangle$ against a laboratory

energy runs through the declared calibration row at one declared inverse temperature, with an affine law, slope and reference uniqueness, and tick conversion; no theorem establishes that the record reference is a Gibbs state, and the calibration carries one declared tick, with two distinct ticks giving two distinct laboratory frequencies.

9.3 Matter

Matter enters as charge carried by the record structure, in two registers the framework keeps distinct. The gauge register is the exterior module of Theorem 8.6: a declared trace-balanced representation whose exterior algebra branches into the fifteen-state chiral generation, with the exhaustive scan selecting exactly one charge-conjugate pair inside the declared menu and the common $\mathbb{Z}_6$ kernel fixing the maximal faithful image. The transport register is integer port load. A write appends $+1$, a retraction appends -1, and at a declared nonzero step h a declared charge hopping along the twelve ports has conserved total charge, satisfies the committed continuity equation identically, and makes the source pairing gauge invariant under endpoint-vanishing gauge functions exactly when continuity holds on the interior steps; its endpoint step difference is the polarization load of the committed coupling at the charge fixing $\kappa = -qh/(12 - 4\varphi)$. Under closed two-step variations of the transported action, exchanging a forward seam crossing with an adjacent rest is stationary exactly when the scaled electric seam field on that seam vanishes there, and along the sourced Ampère evolutions the delay difference equals h times the field-energy transfer at the hop. The hopping sources are consistent with the field sector: the Gauss constraint propagates under the sourced Ampère step exactly when the sources are conserved, and a Gauss-consistent history exists if and only if the initial load is neutral and the sources are conserved, so on the closed carrier one hopping charge alone is admissible only at zero charge and a join carries a neutral partner or a neutralising background (`Lean/Screen/SeamChargeContinuity.lean`). The worldline class and the potentials are declared.

The neutral-partner branch uses one finite action: two declared seam-step worldlines carry charges $q, -q$ and couple minimally to the same Maxwell window functional, with declared clock terms. Endpoint-vanishing gauge transformations preserve the action, its sources obey continuity, and field variation gives the sourced Ampère and Gauss equations. At $q = 1$, $h = 1/2$, an exact rational history on the window $N = 1$ is stationary in both field slots and under every admissible in-window closed two-step replacement of either path, while the Coulomb-started history is field-stationary and fails those exchanges. This is a same-action existence result, not a minimum, stability or dynamical-selection theorem; the paths, action, preparation and clock are declared (`Lean/Screen/NeutralPairCoupledAction.lean`, `Lean/Screen/NeutralPairJointStationaryWitness.lean`). A supplied lattice action also has a conserved fermion current, and a separate quantum link satisfies abelian Gauss; both assume canonical anticommutation relations.

Exchange statistics and exclusion. The internal exterior algebra is not a derivation of physical Pauli exclusion. A supplied spin lift admits both symmetric and antisymmetric two-particle sectors; channel-preserving one-particle-per-channel current histories cannot distinguish their statistics. The gauge analysis gives a conditional exclusion theorem with a precise operational hypothesis [13]. On a finite positive Hilbert space let $C(v)$ be a complex-linear creation family and $N_v = C(v)C(v)^\dagger$. If, for every unit mode,

$$0 \le N_v \le I, \qquad [N_v, C(v)] = C(v),$$

then $C(v)^2 = 0$: in an eigenbasis of N_v, a nonzero entry of $C(v)$ can only raise eigenvalue zero to one. Linearity then gives $\{C(v), C(w)\} = 0$. Full mixed canonical anticommutation additionally

follows if $C(v)$ and its adjoint are the exhaustive Kraus operators of a normalized add/remove instrument.

Identifying $C(v)^\dagger$ with an unscaled Kraus operator supplies the contraction bound, and under the exact phase law square-zero conversely makes $C(v)$ a partial isometry, so within this class the bound is equivalent to exclusion. Finite capacity, covariance and stability alone do not supply it: symmetric Fock space capped at total occupation two preserves the supplied one-particle dynamics and phase law while permitting double occupation. The theorem is an operator criterion; the native field/operation identification and the physical spin–statistics attachment remain underived.

9.4 The action, and why it is stationary

Given a pointwise-continuous one-parameter group of record-preserving star-automorphisms on the finite private algebra, every such group is blockwise unitary conjugation generated by one time-independent self-adjoint Hamiltonian per Wedderburn block, unique up to a real scalar [10, 1]. The Schrödinger form is a theorem about that supplied flow; no theorem selects the flow parameter as physical time or selects one flow from the record alone.

Once the kernel and reference are supplied, the finite path action is fixed rather than independently posited. For any strictly positive row-stochastic kernel and strictly positive normalized initial law, the Markov path law is the exponential tilt of the step-uniform reference by the log-transition action at multiplier one, and an action-multiplier pair reproduces the law exactly when the multiplier-weighted action equals the log-transition action plus a constant; the action is therefore unique up to an additive constant and a multiplier rescaling, and the step-uniform reference is the unique reference invariant under independent target scrambling at fixed source. The declared object is the reference measure; given it, the dynamics selects its own action, and the committed source chain instantiates both statements through kernel-decided integer receipts. Single-site minimality of that action is then a statement about probability: single-site minimizers of the log-transition local action coincide with most probable paths of the realized chain through an exact corner identity, so single-site least action and local maximum likelihood are two readouts of one functional, while an explicit concave stationary history is nonminimal, so stationarity alone is weaker. A finite Legendre bridge joins the two faces: the discrete Euler–Lagrange condition at a junction holds exactly when one step of the discrete Hamilton flow carries the incoming junction state to the outgoing one, for the quadratic class and for strictly convex Lagrangians with a solver section, and the constant Noether current of the chain face equals the quadratic Legendre momentum. One machine-checked theorem composes them: a fixed-endpoint single-site-minimizing embedded history realizes the derived action of the kernel whose exponential tilt is the path law, is an interior most-probable update of that law, and satisfies the discrete Hamilton equations.

Boundary 9.2. The reverse direction is an exact non-identifiability theorem. The bilinear real extension of the source corner table is velocity-affine with no global Legendre solver, while every $L_a = L_0 + \frac{a}{2}y(y-1)$ with $a > 0$ agrees on every realized history yet is strictly convex with an explicit Hamiltonian, so no unique real Hamiltonian continuation follows from this finite history law without an additional curvature receipt. Inside the declared one-parameter family, stationarity of the constant-one transition-weight maximizer under every fixed-endpoint single-site variation forces $a_\star = 2\log(W_{11}^2/(W_{10}W_{01}))$, twice the log of the interior mode-dominance ratio. The selection is grammar-bound: velocity-only quartic continuations share corner invisibility, junction stationarity, the single-site minimum and momentum invertibility, and only a fourth clause of second-order boost null covariance pins the committed point inside the full polynomial grammar, every clause proved load-bearing.

9.5 The field sector on the screen

The same carrier supports an exact static field theory. The seam Laplacian has an explicit symmetric rational Green matrix whose kernel is exactly the constants, every neutral load has a canonical Coulomb solution that is the unique cycle-orthogonal and unique minimal-energy solution of the discrete Gauss problem, and the committed uniform seam repair is one minus one sixtieth of that same Laplacian, so the repair operator and the field operator are the same object up to normalization. The $U(1)$-valued seam connections modulo port gauge transformations are classified exactly by their nineteen chord holonomies, the first cohomology of the thirty-seam graph with circle coefficients being $U(1)^{19}$, matching the source-free Gauss cycle rank. The twenty oriented faces give a genuinely local curvature CA, with C of rank nineteen, kernel exactly the port gradients, and five nonzero entries per seam row in $C^{\mathsf{T}}C$. The local quadratic action is gauge invariant exactly for conserved seam currents, its stationary points exist exactly for those currents, and they are global minima unique modulo gauge.

The complete seam-current symbol also carries a quantitative continuum field theorem. On a supplied common Euclidean domain, set $\omega_a(k) = \sqrt{\Lambda_a(k)}$, and use the opposite-sign curl pair $\mathcal{G}_a(E, B) = (D_a B, -D_a E)$ with $D_a = i\omega_a(k)|k|^{-1}k\times$ at nonzero momentum and zero at the origin. The exact source second and fourth moments imply the global bounds

$$0 \le |k|^2 - \Lambda_a(k) \le \frac{a^2|k|^4}{20}, \qquad 0 \le |k| - \omega_a(k) \le \frac{a^2|k|^3}{20}.$$

Its Fourier propagator assembles to a unitary group on real $L^2(\mathbb{R}^3;\mathbb{R}^6)$: opposite momenta obey the conjugate reality condition, and longitudinal components are fixed. For the local Maxwell group U_0, Plancherel gives

$$\sup_{|t|\le T} \|(U_a(t) - U_0(t))F\|_2 \le \frac{Ta^2}{20}\, \||D|^3 F\|_2.$$

For arbitrary finite-energy data the convergence is strong and uniform on bounded time intervals. Duhamel's formula extends it to the same prescribed current on every scale; charge continuity propagates the Gauss constraint. The global scalar estimates are kernel-checked in `Lean/Screen/SeamMaxwellContinuum.lean`; the real-field, strong-limit, and forcing arguments are proved in the detailed microphysics paper [8]. This is continuum Maxwell dynamics for a declared field family. The Euclidean domain, reversible evolution, common time and physical current identification are supplied, and the finite-scale curl is generally nonlocal. A map from authenticated source histories and decoded observer fields into this family is not constructed.

Boundary 9.3. A finite packet adds a typed continuity equivalence conditional on the declared unit-step Ampère recurrence, whose Faraday identity, gauge invariance, static Coulomb join and staggered quadratic-form balance are exact. For nonzero h, every zero-current datum has uniformly bounded electric seam energy exactly when $h^2(3+\sqrt{5}) < 4$; inside that window the magnetic face energy is also bounded, at and above it an explicit zero-current solution has unbounded electric energy, and the committed unit step lies above the window. Inside it, each eigenvector of the local operator with eigenvalue λ evolves on the amplitude–velocity plane by a determinant-one symplectic map conjugate to rotation by $\arccos(1 - h^2\lambda/2)$, embedded in a continuous one-parameter group that preserves the staggered energy; that energy is positive definite exactly on the window, its step-invariant symmetric forms are exactly its real multiples, and each mode is a complex line with an exhibited phase generator. Nineteen kernel-checked seam vectors over $\mathbb{Z}[\varphi]$, at eigenvalues two, three, five and the golden pair, span the curl sector, the gradient modes form the radical, and every zero-current temporal-gauge solution is the assembled history of a coefficient state (`Lean/Screen/Fi`

`eldSectorEnergyInnerProduct.lean`, `Lean/Screen/CurlSectorEigenbasis.lean`). A principal-angle interpolation places the energy-weighted coefficient vector on a separately constructed $M_{19}(\mathbb{C})$ block with a self-adjoint diagonal Hamiltonian whose rank-one intertwiner has trace equal to the field energy (`Lean/Screen/CurlStoneClockBridge.lean`); this is not an algebra isomorphism with the private carrier, and reading the group parameter as physical time is the declared clock premise. These modules identify no source-selected public representation or Hamiltonian, physical photon Hilbert space, laboratory Schrödinger evolution, source-produced charge or current, calibrated clock, or massless physical particle.

9.6 Coupled matter, continuum trajectories and quantum states

On the supplied solid twenty-tetrahedron cone, exact geometric assembly constructs positive Whitney mass forms and proves $K < 24M$ in Lean; the result supplies the thirty-mode classical and polynomial-quantum consumers. A declared charged-scalar/Maxwell action on that geometry supports a nonlinear continuum estimate and a finite-mesh interacting Hilbert space [10]. Its real, zero-current sector is invariant under the full coupled equations, including terms from the gauge-dependent scalar interpolation. Zero electromagnetic fields give $u_{tt} - \Delta u + m^2 u + g u^3 = 0$. For $m^2 > 0$, $g \geq 0$, a smooth $C_t^2 H_x^2$ Neumann solution and Ritz-projected initial data, a nested conforming refinement, $n = 2^k$, with $20n^3$ cells and uniform shape bounds gives

$$\sup_{0 \leq t \leq T} \left(\|u_n - u\|_{H^1} + \|\dot{u}_n - \dot{u}\|_{L^2} \right) \leq C_T / n.$$

For a smooth magnetic-Neumann reference and magnetic Ritz data, complex matter has the same rate in a prescribed uniform magnetic field. Self-consistent Maxwell backreaction requires a separate estimate.

On the fixed mesh with unwrapped edge coordinates and $e \neq 0$, eliminating the twelve mean-zero gauge directions using the full kinetic form gives a complete metric γ on $\mathcal{Q} = \mathbb{R}^{30} \times \mathbb{C}^{13}$. For supplied $\hbar > 0$ and Laplace–Beltrami quantization,

$$\widehat{H} = -\frac{\hbar^2}{2} \Delta_\gamma + V \quad \text{on} \quad L^2(\mathcal{Q}, d\mathrm{vol}_\gamma)$$

is essentially self-adjoint on compactly supported smooth functions. The nonzero subspace invariant under the remaining constant matter phase reduces its closure and carries unitary evolution. Gauge-invariant scalar-density and magnetic smearings are the same configuration functions classically and quantum mechanically: their maximal multiplication operators are self-adjoint on the neutral sector and have a joint spectral probability law. In the real sector, bounded Lipschitz density readouts inherit the $O(1/n)$ trajectory rate. Quantum continuum convergence and selection of quantization are separate requirements.

Five self-reading patches evolve field records through feedback. Interval integration and authenticated replay bound 81 decoded position/velocity checkpoints within 1.0001×10^{-10} of the charged solution on $[0, 2]$, whose lift satisfies 68 Euler–Lagrange and 13 Gauss equations. For regular paths in $E > V$, the temporal-gauge duration $d\tau = \sqrt{G_q(dq, dq)/(2(E - V(q)))}$ is reparameterization invariant; on nonturning solutions of energy E it recovers action time. Geometry, couplings, preparation, quantization and laboratory identification are inputs [10, 1].

9.7 Standard Model action structures

Table 3 distinguishes gauge and fermion forms constrained by the finite representation, scalar and Yukawa interactions supplied as inputs, and parameters not determined by the construction. It records action shapes and relative structures; numerical coupling values are separate inputs.

Table 3: The canonical Standard Model term-and-sector inventory, classified by what the reconstruction supplies.

Term or sector	**Classification**
SU(3), SU(2), and U(1) gauge kinetic terms	partial
Nonabelian gauge self-interactions	partial
Quark doublet, up-singlet, down-singlet, lepton doublet, and electron-singlet kinetic and covariant terms	partial
Right-handed neutrino stance	derived conditional
Higgs kinetic and covariant term	registered premise
Higgs potential	registered premise
Up-type, down-type, and lepton Yukawa terms	registered premise
Theta-QCD term	absent
Generation triplication	partial
Gauge couplings g_1, g_2, g_3 as numerical parameters	absent
Quark and lepton mixing structure	partial

The four classifications mean different things. The derived conditional row is the right-handed neutrino stance: within the declared grammar the exhaustive scan admits exactly one unordered charge-conjugate pair of chiral anomaly-free packages, each with fifteen Weyl states and no right-handed neutrino, with neutral anomaly-free singlets outside the scanned space. The partial rows carry an attained shape without a source-produced action: ad-invariance over each certified compact bracket forces a block-diagonal kinetic form with one free positive coefficient per simple factor, the Yang–Mills shape, and the registered exterior grammar fixes each fermion line's covariant-derivative coupling pattern through its hypercharge. The registered-premise rows are consumed as declared: no theorem produces the scalar doublet, its potential or coefficients, the vacuum, the electroweak scale, or any Yukawa matrix, and an exact non-definability result leaves a free positive rescaling family of admissible up-type spectra. The two absent rows have no structure in the framework.

Conditional flavor calculations constrain these missing inputs. Complete uniform-reference entropy on the six antipodal pair records has an axial minimum at every positive fixed amplitude, so its single-quadrupole readout has a repeated eigenvalue. A specified append-and-repair protocol instead fixes exact stationary weights on 6077 relative settled states, with mean repair cost 11/24; its rotation-invariant average supplies no family hierarchy. Exact Higgs contractions and kinetic normalization show that a common record matrix in both kinetic and Yukawa terms cancels from the normalized spectrum [9].

The target-informed six-mass formula can be replayed at independently fixed $P = \varphi + \sqrt{\pi}\alpha$ under its declared mass charts, but its favorable individual comparisons coexist with a failed light-mass ratio. A separate top-only source attachment to the Landau-gauge modified-minimal-subtraction vacuum supplies finite electroweak and complex-pole matching with every declared scale and order control retained. The particle paper gives the mass conventions, numerical tables and electroweak disagreements [9]. This conditional calculation does not select the vacuum or gauge attachment, and the zero light Yukawas do not yield six nonzero physical masses.

Two exact discriminators constrain the gauge sector without selecting it. A declared equal-weight cyclic rule on the twenty oriented faces gives the A_5-equivariant bracket $B_{\mathrm{face}} = 60R_{13}$, which fails Jacobi in exactly 240 of 2640 independent coordinates, and exact primal–dual certificates place the family G nearest to it in the classified compact locus under the ℓ_1, ℓ_2^2 and ℓ_∞ coordinate edit norms; independently, ad-invariance reduces the F/G three-weight space to dimension two through $F : w_{3+} = \sqrt{5}\, w_5$ and $G : w_{3-} = \sqrt{5}\, w_5$, while the P control stays two-dimensional [13]. These are representation-level statements over committed finite packets; the relative coefficients

between simple factors, the abelian kinetic ray, and the matter, scalar and hypercharge sectors of any composed effective action require the prospectively fixed source-selection protocol.

10 Exact quantitative closures and physical tests

The finite architecture supports a small set of quantitative surfaces sharp enough to be tested numerically. Each surface below is classified by Table 1: the theorems are finite theorems, the numerical enclosures are certified finite computations, and every empirical surface is typed as a diagnostic, a conditional empirical test, or a frozen prospective branch prediction. The closure hypothesis of Section 1.5 contributes its quantitative candidates here, in Section 10.1, under the same typing.

10.1 Closure candidates: the screen grain and the capacity coordinate

Two declared map candidates motivated by the closure hypothesis are examined, one for the screen grain and one for the record capacity, and both carry interval-certified roots. Neither clears the source-only selection and same-quantity gates. The screen-grain loop

$$P = \varphi + \sqrt{\pi}/A_T(P)$$

has one interval-certified fixed point per declared map, where $A_T(P)$ is the inverse coupling the declared map returns at trial grain P and $\varphi = (1+\sqrt{5})/2$. The bare source map, with the unified gauge width absent, settles at $\alpha^{-1} = 136.994835177413$, a relative distance of 3.0×10^{-4} from the 2022 CODATA value 137.035999177(21) [80]; the self-consistent map, which recomputes the finite-screen unified gauge width inside the loop, settles at $P = 1.6309682414$ and $\alpha^{-1} = 137.035660136947$, a relative distance of 2.5×10^{-6}, the gauge width supplying 99.2 percent of the gap between the bare root and the measured value. The residual 3.3904×10^{-4} exceeds the experimental standard uncertainty by about 1.6×10^4 standard uncertainties, and its one declared address, same-scheme low-energy transport to the Thomson endpoint with hadronic vacuum polarization, is computed by neither map [12]. The capacity branch gives the candidates 3.2921×10^{122} and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck base-ΛCDM chain, residuals of -0.63 and -0.39 percent. Both comparisons are diagnostics of declared maps in the vocabulary of Table 1, exposed before the branch choice, so neither carries predictive weight; the choice of map is itself declared. The screen-grain comparisons comprise two interval-certified map roots and a separate approximate compressed-code output, without a certificate for the latter's fixed point. A separate retrospective substitution grid contains 48 labels for 42 distinct constant pairs, including the declared pair and 41 alternatives; none of those alternatives reaches the declared 2.5×10^{-6} relative-distance threshold. This grid does not exhaust map-form choices. Counts of tried maps or substituted constants specify selection exposure; they supply no chance probability without a declared sampling measure and selection rule. A physical endpoint additionally requires target-independent selection of the map, a typed bridge identifying the two sides as readings of one quantity, and, for the coupling, the same-scheme hadronic transport.

The capacity coordinate carries the cosmological constant through one dictionary. Reading the record capacity as the de Sitter horizon entropy in nats gives

$$N = \frac{3\pi}{\Lambda\, \ell_P^2}, \tag{15}$$

with $\ell_P^2 = \hbar G/c^3$, so the two capacity candidates read $\Lambda = 1.0959 \times 10^{-52}$ and $1.0933 \times 10^{-52}\,\mathrm{m}^{-2}$ against $3\Omega_\Lambda H_0^2/c^2 = 1.0891 \times 10^{-52}\,\mathrm{m}^{-2}$ at the Planck base-ΛCDM centrals, distances of $+0.62$

and +0.38 percent. Equation (15) is the same dictionary that produces the comparison coordinate, so this is the capacity diagnostic expressed in the constant rather than an independent comparison, and it consumes the measured $\hbar$, G and c through the calibration import (M2). Large capacity explains the small dimensional constant, not its agreement with the comparison: a relative capacity residual δ gives a relative constant residual $-\delta/(1+\delta)$.

A covariance-preserving transform of the four official DESI DR2 base-ΛCDM chains for BAO+CMB, each authenticated against its manifest, gives the retrospective coordinate $\Lambda\ell_P^2 = (2.96770 \pm 0.03978) \times 10^{-122}$ [79, 1], against which the two capacity candidates read 2.86285×10^{-122} and 2.85591×10^{-122}, residuals of -3.53 and -3.77 percent at -2.64 and -2.81 posterior standard deviations. The Planck centrals quoted above carry no attached uncertainty, so the two displays are comparisons against different objects and disagree in sign; neither is an OPH likelihood, theory uncertainty or predictive pull.

10.2 The positive-chamber Koide identity

The icosahedral carrier's face structure supplies order-three cyclic response operators. The following exact identity governs every Hermitian three-cycle response [14].

Theorem 10.1 (Positive-chamber Koide identity)**.** *Let $R^3 = I$ generate a cyclic action and let*

$$C = aI + bR + \bar{b}R^2, \qquad a > 0, \quad b = \rho e^{i\delta} \in \mathbb{C},$$

be the associated Hermitian circulant, with eigenvalues

$$\lambda_k = a + 2\rho \cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

Suppose all three eigenvalues are nonnegative and, for one scale $s > 0$, $\sqrt{m_k} = \sqrt{s}\,\lambda_k$. Then

$$Q := \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{\rho}{a}\right)^2, \qquad \text{so} \qquad Q = \frac{2}{3} \iff \frac{\rho}{a} = \frac{1}{\sqrt{2}}. \tag{16}$$

Proof. The three cosines at angles separated by $2\pi/3$ sum to zero and their squares sum to $3/2$, so $\sum_k \lambda_k = 3a$ and $\sum_k \lambda_k^2 = 3a^2 + 6\rho^2$. With $m_k = s\lambda_k^2$ the ratio is $Q = (3a^2+6\rho^2)/(3a)^2 = \frac{1}{3} + \frac{2}{3}(\rho/a)^2$, and solving $Q = 2/3$ on $a > 0$, $\rho \geq 0$ gives the equivalence. □

The shape depends on ρ/a and δ. Fixing $Q = 2/3$ fixes ρ/a and leaves one phase: the two mass ratios vary together. Equal rank-two event blocks in the finite tracial Gelfand–Naimark–Segal packet supply the balance $\rho/a = 1/\sqrt{2}$ exactly under the declared tracial premises, and the Lean development checks both the circulant identity and the equivalence [14].

Under the balance and mass-ordering premises the measured electron and muon masses [81] determine the tau mass through one quadratic, governed by a prospectively fixed conditional test whose target definition, precision floor of 0.045 MeV and decision rule were fixed before comparison with any post-registration measurement [3]. The outward-rounded enclosure is [1776.968991, 1776.969063] MeV, a 72-eV window whose center sits 0.43σ from the measured 1776.93 ± 0.09 MeV. The balance premise is the equal-weight condition on the two response sectors of one icosahedral face circulant, and it reproduces the relation Koide published for the measured charged-lepton masses, so comparisons with the measured triple are postdictions; the registered tau window carries the forward test, and a world-average or dedicated central value more than three standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise and with it the equal-rank-two-block tracial reading of the face-circulant sector. Alternate family-sector carriers, potentials, dynamics and source attachments are not excluded, and the surface supplies no physical family attachment or phase selection.

10.3 Finite capacity and the de Sitter transfer sign

The screen's finite capacity supports an exact entropy accounting with one conditional gravitational reading [15].

Proposition 10.2 (Capacity transfer sign)**.** *For finite sector dimensions d_i with total $M = \sum_i d_i$ and sector probabilities p_i, the generalized entropy obeys the exact identity*

$$S_{\mathrm{gen}}(p, d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i = \log M - D\Big(p \,\Big\|\, \frac{d}{M}\Big),$$

where D is relative entropy; its exact maximum is $S_{\mathrm{gen}}^{\max} = \log M$ at $p_i = d_i/M$. If an observer receives a fraction $f \in (0, 1)$ of a fixed total capacity and the sectors deplete uniformly, every admissible transfer changes the maximum by

$$\Delta S_{\mathrm{gen}}^{\max} = \log(1 - f) < 0,$$

and the positive-real interpolation in f is strictly decreasing and concave, so the zero-transfer point is a one-sided boundary maximum.

Proof. Expanding $D(p\|d/M)$ gives the identity, and nonnegativity of relative entropy with equality at $p = d/M$ gives the maximum. Uniform depletion replaces M by $(1 - f)M$, so the maximum changes by $\log(1 - f)$, whose monotonicity and concavity are elementary. □

Identifying capacity with horizon area, the transfer coordinate with observer mass, and the associated finite port operator with the gravitational shock supplies a conditional de Sitter time-advance mechanism with a definite sign: under those dictionaries an observer absorbing a capacity fraction f strictly lowers the maximal generalized horizon entropy by $\log(1 - f)$, and the one-sided boundary maximum carries the definite sign of the associated time shift. The finite identities make none of those physical identifications; no horizon or observer dictionary is supplied, and the finite calculation supplies no positive cyclic static-patch trace.

10.4 The unique rank-six invariant of the primitive carrier

In this subsection, "spin six" means spherical-harmonic angular rank $j = 6$. It does not denote a particle with spin six. The mathematical engine is the invariant count for the sixty-element proper icosahedral rotation group. The dimension of the invariant subspace in angular rank L is

$$m_L = \frac{1}{60}\Big[(2L + 1) + 15\,U_{2L}(0) + 20\,U_{2L}(\tfrac{1}{2}) + 12\,U_{2L}(\tfrac{\varphi}{2}) + 12\,U_{2L}(\tfrac{\varphi^{-1}}{2})\Big],$$

where U denotes the Chebyshev polynomials of the second kind. The exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$. Thus no nonconstant invariant occurs below angular rank six, and the rank-six invariant line is one-dimensional, in agreement with the classical icosahedral-harmonic classification [59].

Theorem 10.3 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2} \sum_d w(d)\,[1 - \cos(a\,k \cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a; every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, whose stationary structure is the exact 62-direction census (12 and 20 extrema of opposite index, 30 saddles); and whenever the weighted sixth-moment $\mathcal{I}_6$ coefficient is*

nonzero, a condition certified for the equal-weight stencil and for each fundamental orbit, the first directional artifact appears at order a^4k^6 and one binary refinement step suppresses it by exactly $1/16$ at that order. A vanishing sixth moment removes the displayed a^4k^6 residue. Any surviving directional term occurs at higher order. For a single-radius member whose weighted sixth-moment coefficient vanishes, the rank-six component vanishes at every order, so any surviving anisotropy has a higher even invariant rank. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $L \le 2m$. The invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$, so every anisotropic term below rank ten is proportional to $\mathcal{I}_6$ and the possible artifact ranks are the even invariant levels $6, 10, 12, 16, \ldots$; the scaling in a is fixed by the expansion order. The invariant table, the 62-direction census, the orbit verification, the nonvanishing sixth-moment coefficients, and an exact positive-weight tuned member whose rank-six content cancels are certified in exact arithmetic [1]. □

The universality theorem concerns a class of finite operators, and a physical prediction requires a narrower branch whose operator and transfer are fixed. For the declared equal-weight twelve-port member the stationary structure is exact at every scale: the 62 critical directions of $\mathcal{I}_6$ define 31 projective axes, and three exact projective charts with a 2,624-leaf interval cover prove that the full cosine kernel has exactly those directions for every $0 < |ak| \le 1$, the twelve port directions remaining maxima, the twenty face-center directions minima, and the thirty edge-center directions saddles [1].

Let $u_1, \ldots, u_{12}$ be the unit vectors through the twelve carrier ports. The *primitive twelve-port propagation branch* declares that the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 10.3 with scale-independent coefficients; that the complete primitive orbit is its sole hop support, with no independent isotropic or directional term through order k^6; that the carrier scale a is finite and positive with the continuum quadratic term normalized to k^2; that one carrier rest frame is transported coherently over the experiment; and that the tested scalar or polarization-independent sector realizes the operator, a photon realization additionally requiring equal action on both transverse polarizations. Transitivity of the proper carrier group forces equal weights once the single orbit is selected. The carrier is realized as a quasiperiodic structure or as a graph, since no periodic three-dimensional lattice carries this point group, so the branch presupposes no crystal.

The finite repair packet leaves this branch avoidable: its certified operator acts on thirty internal seam readings, no committed artifact joins a complete twelve-port translation operator to a physical readout of that operator, the equal-weight vertex, face and edge stencils have pairwise distinct rank-six rays, and proper-carrier transitivity selects no orbit as the physical hop support.

Proposition 10.4 (Primitive twelve-port branch prediction)**.** *Under the primitive-port propagation premises, the physical symbol is*

$$\omega^2(k, \widehat{k}) = \frac{1}{2a^2} \sum_{i=1}^{12} \Big[1 - \cos\Big(ak\, u_i \cdot \widehat{k}\Big)\Big]. \tag{17}$$

With

$$\mathcal{I}_6(n) = \frac{25}{132} \sum_{i=1}^{12} P_6(u_i \cdot n), \qquad \mathcal{I}_6(u_i) = 1,$$

where P_6 is the degree-six Legendre polynomial. The small-ak expansion is

$$\omega^2 = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8). \tag{18}$$

In a transported frame, $\omega^2 = k^2 + C_4 k^4 + B_0 k^6 + B_6 k^6 \mathcal{I}_6(R^{-1}\widehat{k}) + O(k^8)$ *with* $R \in \mathrm{SO}(3)/A_5$*, so the branch predicts* $C_4 = -a^2/20$*,* $B_0 = a^4/840$ *and* $B_6 = 2a^4/7875$*, and hence the scale-free relations*

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}. \tag{19}$$

All intrinsic anisotropic coefficients at $1 \le j \le 5$ *vanish. Once a negative* C_4 *is resolved on this branch, no scale or anisotropic amplitude is free; only one three-parameter orientation class* R *in* $\mathrm{SO}(3)/A_5$ *is profiled. Under binary refinement,* $C_4(a/2) = C_4(a)/4$ *and* $B_0(a/2) = B_0(a)/16$*,* $B_6(a/2) = B_6(a)/16$ *at fixed physical momentum.*

Proof. The exact carrier moments $\sum_i (u_i \cdot n)^2 = 4$, $\sum_i (u_i \cdot n)^4 = 12/5$ and $\sum_i (u_i \cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$, substituted into the cosine series of (17), give (18); eliminating a gives (19), and the lower-rank nulls and uniqueness of the rank-six line follow from Theorem 10.3. □

Proposition 10.5 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module* D_6 *of Theorem 2.3, with Smith invariants* $(1,1,1,1,1,2)$*, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight* $1/60$ *on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2*, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then*

$$\Lambda_a(k, \widehat{k}) = \frac{1}{5a^2} \sum_{j=1}^{30} \left[1 - \cos\left(ak\, w_j \cdot \widehat{k}\right)\right], \tag{20}$$

with $C_4 = -a^2/20$*,* $B_0 = a^4/840$ *and* $B_6 = -a^4/12600$*, hence* $B_0/C_4^2 = 10/21$*,* $B_6/C_4^2 = -2/63$ *and* $B_6/B_0 = -1/15$*. The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated* $\mathcal{I}_6$ *orbit.*

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 2.3. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j \cdot n)^2 = 10$, $\sum_j (w_j \cdot n)^4 = 6$ and $\sum_j (w_j \cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (20) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The declared symbol also carries an exact remainder certificate with no physical premise: with $q = ak$, $|\widehat{\Lambda} - P_6| \le \frac{7}{388800} q^8$ and $\frac{19}{20} q^2 \le \widehat{\Lambda} \le q^2$ uniformly on $0 \le q \le 1$, where $P_6 = q^2 - q^4/20 + (1/840 - \mathcal{I}_6/12600) q^6$; the certificate turns a into no SI length. The

edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

One exact theorem binds the two frozen rays and every member of the declared positive-weight scalar cosine class into a single test surface.

Theorem 10.6 (Carrier-class dispersion band)**.** *Consider the declared positive-weight scalar cosine class of spatial symbols*

$$\Lambda(k,\widehat{k}) = \sum_s w_s \sum_{u\in O_s} [1-\cos(ak\, r_s\, u\cdot\widehat{k})],$$

where $a>0$, each O_s is one orbit of the sixty proper carrier rotations on unit directions, the finitely many shells carry radii $r_s>0$ and weights $w_s>0$, the displayed cosine sum is the full spatial symbol and hence complete through order k^8, and the continuum normalization fixes the k^2 coefficient to one. Write the normalized expansion as

$$\Lambda_{\mathrm{norm}} = k^2 + C_4k^4 + (B_0 + B_6\mathcal{I}_6(\widehat{k}))k^6 + (D_0 + D_6\mathcal{I}_6(\widehat{k}))k^8 + O(k^{10})$$

and set $\mu_m = \sum_s w_s|O_s|r_s^m$. Then five exact statements hold.

1. *Sign law: $C_4 = -(a^2/20)\,\mu_4/\mu_2 < 0$ for every member.*
2. *Isotropic floor: $B_0/C_4^2 = (10/21)\,\mu_2\mu_6/\mu_4^2 \geq 10/21$, with equality exactly on the single-radius members.*
3. *Rank purity: the intrinsic anisotropic ranks one through five vanish and the rank-six residue is one multiple of the rotated $\mathcal{I}_6$.*
4. *Rank-six band: every member has $B_6/B_0 = (16/75)\,\langle\mathcal{I}_6(\widehat{u}_s)\rangle$, the mean taken with the positive weights $w_s|O_s|r_s^6$, and this ratio lies in $[-16/135,\ 16/75]$ for the declared class. The pure face and vertex orbits attain the endpoints, the pure edge orbit sits at $-1/15$, and the vertex-face mixture at per-direction weight ratio 25:27 attains zero.*
5. *Eighth-order confinement and common-radius lock: every member has*

$$D_0 = -\frac{a^6}{60480}\frac{\mu_8}{\mu_2}, \qquad \frac{D_6}{D_0} = \frac{64}{125}\,\langle\mathcal{I}_6(\widehat{u}_s)\rangle_{w_s|O_s|r_s^8}.$$

No independent angular shape occurs at order k^8. Every member whose active shells share one radius obeys the division-free identity $5D_6B_0 = 12B_6D_0$. This includes the 25:27 zero-anisotropy mixture, where both sides vanish.

Proof. Summing over the sixty proper rotations gives, on unit vectors,

$$\sum_g \big((gu)\cdot\widehat{k}\big)^6 = \tfrac{60}{7} + \tfrac{64}{35}\,\mathcal{I}_6(\widehat{u})\,\mathcal{I}_6(\widehat{k}), \qquad \sum_g \big((gu)\cdot\widehat{k}\big)^8 = \tfrac{20}{3} + \tfrac{256}{75}\,\mathcal{I}_6(\widehat{u})\,\mathcal{I}_6(\widehat{k}),$$

since the invariant multiplicities $(1,0,0,0,0,0,1)$ through rank six and $(1,0,0,1,0)$ at degree eight pin the invariant spaces to the radial power and $\mathcal{I}_6$. The moment ratios give C_4 and B_0; the floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_iW_jr_i^2r_j^2(r_i^2-r_j^2)^2 \geq 0$ with $W_s = w_s|O_s|$, vanishing exactly at one radius; the band follows because the rank-six numerator is a positive-weighted mean of seed $\mathcal{I}_6$ values inside the census range $[-5/9, 1]$; the degree-eight kernel gives D_0 and the weighted mean; and at one common radius $(64/125)/(16/75) = 12/5$ proves the lock. A second series route and a high-precision fit reproduce the table, and a signed-weight control violates the floor [1]. □

The frozen vertex and edge branches sit at $16/75$ and $-1/15$ of the band. The observable quartic length is $a_{\mathrm{eff}} = a\sqrt{\mu_4/\mu_2} = \sqrt{-20C_4}$. A quartic bound constrains this effective carrier length; a bound on a requires independent control of μ_4/μ_2 (the fixed unit-radius branches have $a_{\mathrm{eff}} = a$). Indeed, $a \mapsto a/c$, $r_s \mapsto cr_s$ leaves every physical displacement and the full symbol unchanged. Under the same physical-sector premises, a resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, or a rank-six residue off the rotated $\mathcal{I}_6$ template, excludes every member of the declared class at once, while exact saturation of the floor certifies common-radius support. The certificate reads no comparison data and changes no frozen bytes [1]. A class-wide null verdict additionally needs a positive physical lower bound on a_{eff} and a power contract.

The edge ray and its complete decision rule are frozen under public commit custody before any eligible edge-branch comparison payload [3], with the template class, every primitive-port comparison input, and every datum inspected before that freeze excluded. A null carries no verdict unless a pre-exposure source theorem for the same action and sector supplies $a_{\min} > 0$ and a preregistered power and remainder contract makes the entire admitted manifold excludable.

Primitive-port prediction and decision rule. Proposition 10.4 is a frozen prospective conditional physical-branch prediction [3]. The minimal locally Lorentz-invariant Standard Model plus General Relativity in local vacuum supplies the baseline $C_4 = B_0 = B_6 = 0$ for intrinsic propagation; nonminimal effective operators, media, curvature, or another icosahedral system can imitate parts of the pattern, so a match distinguishes the branch from that baseline without identifying OPH uniquely. In the Standard-Model Extension coefficient language [77, 78] the branch populates only the thirteen-component nonbirefringent rank-six multiplet at operator dimension eight, collapsed to one amplitude and one frame orientation, with every anisotropic coefficient at $1 \le j \le 5$ exactly zero in the carrier rest frame. The exposure class is fixed: the dated template search, whose icosahedral rank-six and higher templates returned a family-wide p-value of 0.64, and every data product examined there are ineligible, and the target statement, exact coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content [3]. A verdict requires a dataset-specific registration fixing one post-freeze release, a joint likelihood or full covariance for the same-sector coefficient vector, the carrier and observer frames, the $\mathrm{SO}(3)/A_5$ orientation profile, the boost law, the nuisance model, trials accounting, sensitivity floor, and calibrated joint threshold. The rule scores the complete branch manifold

$$C_4 < 0, \qquad B_0 = \frac{10}{21}C_4^2, \qquad B_6 = \frac{32}{315}C_4^2,$$

with every intrinsic coefficient at $1 \le j \le 5$ zero and the $j = 6$ vector on the rotated $\mathcal{I}_6$ orbit. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, an isolated lower-rank coefficient is nonzero, the linked sixth-order terms are excluded at adequate sensitivity, or the calibrated profile likelihood excludes the complete manifold. Support requires exclusion of the zero-coefficient baseline at five standard deviations or more, agreement with the linked manifold within two, rejection of named systematic alternatives, and an independent replication. A null is inconclusive, since the branch supplies no positive lower bound on a. A failed verdict rejects the primitive twelve-port physical propagation branch, and applies to the framework as a whole only after a stronger source law makes that branch forced and exclusive.

Boundary 10.7. No statement in this section is a source-only determination of a physical constant or mass. The Koide identity and capacity identities are finite theorems; the tau window is a prospectively fixed conditional test, postdictive in premise ancestry; the lepton enclosures and the

screen-grain separation are diagnostics that consume measured inputs; the de Sitter reading is conditional on unconstructed dictionaries. The angular-rank theorem is a finite theorem. The primitive-port and source-seam coefficient relations form separate prospective predictions on their named physical branches, with empirical scoring sealed until the required branch-specific contracts exist.

11 Physical interpretation maps and their consequences

The preceding finite results become physical claims only when one carrier realizes Definition 1.2 and the standing maps of Section 1.3. Quantum measurement requires actual durable records, protected boundaries and the algebra-state map (M2); event geometry requires one source-selected refinement family with order embedding, calibrated density, manifoldlikeness, topology and convergence (M1, M4). The finite capture does not supply those physical hypotheses. A realized null balance, Ward conservation, the Bianchi identity and an independent scale yield Theorem 7.2 (M2, M4). One carrier may feed both the event/modular and twelve-port response branches, but equality of G_{packet} and G_{Tan} remains a separate identification (M3, M4). Complete response and endogenous transport fix the compact local Lie type; laboratory currents, quantum fields, families and masses require their physical maps (M3, M5). The following sections state the corresponding dark-sector, measurement and thermodynamic consequences. None turns the finite relaxation generator into a physical Hamiltonian or the dispersion symbol into a photon law without the named reconstruction, sector and clock premises (M1, M5).

11.1 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

The two dark components enter the physical Einstein branch only through separate typed adapters: a dark-sector adapter identifies non-luminous modular charge, its stress and the absence of an electromagnetic readout, and an independent horizon–record adapter identifies a selected finite capacity with de Sitter entropy and the vacuum metric coordinate. Neither is part of the minimal finite Einstein-shape theorem, and the companion papers carry the full conditional statements [16, 15, 17]. Under the supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge gives a conditional classification identity: the rest-frame Einstein relation responds to total modular charge, the luminous-only relation holds exactly when the non-luminous charge vanishes, and a geometric excess over the luminous term is classified as non-luminous modular charge, without deriving that a nonzero excess exists or its abundance. A collar-recovery and modular-stress adapter supplies one candidate carrier: anomalous modular energy on overlap collars, bounded by the collar recovery defect, absent at full recovery, diluting as a^{-3} under the adapter's conservation hypothesis, and asymptotically suppressed as density or depth grows for common envelope parameters; source existence, abundance and galactic localization remain separate premises.

Conditional on two named premises, the deep-regime profile is characterized rather than supplied. An enclosed-mass law that is degree-one scale covariant in the radius and composes independent sources in quadrature, because variances of independent collar defects add, carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$; this forces the deep anomalous acceleration $a_A = \sqrt{a_b a_0}$, flat rotation curves, and the baryonic Tully–Fisher relation $v^4 = GM_b a_0$, with each premise proved load-bearing by an explicit counterexample law. A per-cut model reduces quadrature composition to variance additivity on nr nested cuts and gives $a_0 = Gn^2c$, with only the

product n^2c identifiable. The corrected SPARC diagnostic compares the total observed acceleration with $a_b+\sqrt{a_b a_0}$; on 97 common galaxies its paired log-ratio interval $[-0.117, 0.026]$ dex contains zero, and a source-bound replay of the committed snapshot recovers the published interpolation scale $a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,\mathrm{m\,s^{-2}}$, which was known before the replay and is fixed by no source law. The value of a_0, the physical attachment of the envelope, a relativistic stress and lensing law, and the abundance are not constructed [16, 1].

The separate rotation-curve profile uses a fixed catalogue data measure, including the inclination Jacobian. Its contours are uncalibrated sensitivity sets, so they cannot accept or reject a source value of a_0.

The vacuum side of the sector is the capacity coordinate of Section 10.1 read as a density. Identify $\rho_{\mathrm{DE}}(a) = \kappa/N(a)$ at constant κ and take the continuity definition $w(a) = -1 - \frac{1}{3}\,\mathrm{d}\ln\rho/\mathrm{d}\ln a$. The two definitions give

$$w(a) = -1 + \frac{1}{3}\frac{\mathrm{d}\ln N}{\mathrm{d}\ln a}, \tag{21}$$

so a fixed record capacity implies $w = -1$ and the exact Chevallier–Polarski–Linder point $(w_0, w_a) = (-1, 0)$, a monotone nondecreasing capacity implies $w \geq -1$ at every epoch, and $w < -1$ implies local capacity loss. The equation is an algebraic consequence of the two definitions and derives neither fixed capacity nor the density identification. The scoring is asymmetric: exclusion of $(-1, 0)$ in any direction can fail the fixed branch, while compatibility with that point supports nothing, since it is also the ΛCDM null, and a w_0w_aCDM posterior is a model- and likelihood-dependent projection whose comparison requires a post-freeze dataset combination, statistic, thresholds and nuisance policy fixed in advance [3].

11.2 A common source, and measurement as a completed record

One observer-consensus tower feeds two readout branches: the overlap, record and repair structure feeds the event and modular branch, while the twelve-port incidence, complete reversible response and endogenous overlap transport feed the local gauge-type branch, with trace-balanced blocks supplying a separate conditional matter branch. That common source identifies gravity with no gauge boson and fixes no coupling, and it leaves G_{Tan}, reconstructed from a sector category, distinct from G_{packet}, the maximal faithful image of the explicit finite response and matter module; a completed unification requires joint physical realization and a proof that the corresponding current diagrams commute. Measurement sits at the transition from a private repairable record to a protected public normal form: a record is public when it has survived every authorized comparison and no repair can change it, event projectors, weights and state updates are defined on the completed algebra-state surface, and a physical application must identify actual durable records and verify the transaction and readback premises.

11.3 Thermodynamics from conditional repair

The four laws form a finite conditional theorem package by short elementary arguments once Axiom 3 is read on transition distributions and the typed source and physical receipts are supplied. Axiom 3 applies separately to states and to transition distributions, with exact projection identities. The exact finite projection is invariant under changes of observable units and identity offsets; numerical optimizer error remains distinct from family closure error. When the conserved-moment surface contains a faithful feasible state, the projection with faithful reference is a finite-multiplier Gibbs exponential state, characterized by the information-projection Pythagorean identity. Boundary constraints can instead force a rank-deficient optimum requiring a support-restricted or limiting

treatment; a faithful reference alone does not exclude this case. On one supplied nondegenerate finite spectrum equality of Gibbs distributions identifies the inverse temperature, which is the finite zeroth-law transitivity receipt rather than an equilibration theorem. The transition projection onto the fibre of the complete repaired visible datum is weighted conditional resampling from the same reference, the weighted observation-fiber projector of the consensus construction. That kernel is stochastic, idempotent, reversible, and stationary, fixes every fibre-measurable charge, and contracts relative entropy to the reference, so the second law is a data-processing theorem with the modular form $\Delta S \geq \Delta\langle K\rangle$ on τ-preserving channels and the Landauer erasure bound as a corollary.

The inequality is the mean of a fluctuating entropy production obeying exact integral, pointwise, and level-set fluctuation identities, and the mean is an identity: one repair step's mean entropy production equals the relative entropy from the input state to its repaired image, strictly positive whenever the step changes a strictly positive state and zero exactly on the fibre-conditional-reference fixed points. The identical kernel is idempotent and produces no further dissipation after the first projection, so a sustained macroscopic arrow is unclaimed. Detailed balance with a linear Poisson solver on centered currents gives a symmetric positive-semidefinite finite Green–Kubo matrix with an exact cutoff remainder [29, 30]; a decaying memory tail with a stabilizing sum requires a separately sourced nonidempotent evolution. For a finite joint update the exact bookkeeping split is $\Delta U = \delta Q + \delta W + \operatorname{Re}\operatorname{tr}(\delta\rho\,\delta H)$, and the composed repair heat stroke holds H fixed. At every finite regulator the excited Gibbs mass is bounded by $\frac{d-g_0}{g_0}e^{-\beta\Delta}$, the entropy limit $k_B \log g_0$ is a finite corollary of that bound, and finite-step unattainability follows because faithful repair and pinching steps cannot reach the rank-deficient zero-temperature state. Through the central split $K = 2\pi B + Z$ one repair step with a fibre-measurable central charge obeys the cap Clausius inequality $2\pi\Delta\langle B\rangle \leq \Delta S$, which discharges the Einstein branch's finite first-law premise package on the simplex tangent space of this model.

Boundary 11.1. The strict-descent normalizer that settles public facts is a different map and carries no entropy inequality, with an explicit certified two-point counterexample. The pinned source artifact has an exact negative verdict: its state-side resampling action is idempotent while its recurrent transition action has a nonconstant eigenmode with $\lambda = 665437/726948 \in (0, 1)$, so its stationary mass is no deterministic pushforward of the state reference. Physical thermodynamics additionally requires source-justified transitions, a shared reference, energy-clock calibration and uniform low-temperature tails on one cofinal family [10, 1].

A constructive finite alternative keeps the full-fibre projector P as equilibrium and uses local heat-bath evolution T. An exact eight-state nonproduct witness preserves a nonconstant public bit, obeys $TP = PT = P$ and the entrywise minorization $T^2 \geq (7/96)P$, and has $\|T^n\| \leq (89/96)^{\lfloor n/2\rfloor}$ on fibre-centred observables, giving a unique centred Poisson solver and convergent Green–Kubo sums. Conserved records thus coexist with decaying memory under this supplied law; native source selection and physical transport are separate [10, 1].

11.4 What fails together

The conditional implications share premises, so they fail in groups. Failure of (M4) removes the effective-spacetime and Einstein conclusions together. Failure of (M3) removes the gauge and matter interpretations together. Failure of (M2) removes every statement in laboratory units, including the mass-energy slope and the four-law package in physical form. The finite normal-form, invariant-theory, and algebraic results retain their stated domains under all three.

12 Machine verification and reproducibility

Each result maps to an artifact class: a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels; exact-arithmetic code receipts, integer and rational computations, outward-rounded interval certificates and $\mathbb{F}_2$ rank computations, each emitting a canonical hashed payload; simulation receipt bundles pinning the source or module revision, configuration, seed where stochastic, grids, tolerances and output hashes, with deterministic reserialization distinguished from fresh replay; verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees, so a caller cannot assert a truth flag directly; adversarial negative controls, so that every certificate ships with mutations that must be detected and a control that cannot fail counts as a defect of the certificate; and provenance rules under which machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test, or rejected candidate and reject any stronger classification unsupported by the recorded dependencies.

The repository, receipts, schemas, and rebuild instructions are public [1, 2]. The formal and executable stack checks the claims assigned to it and enforces their stated boundaries. Analytic arguments in this manuscript retain their displayed mathematical proofs, and no part of the stack establishes physical realization.

12.1 Artifact map

Table 4 names the artifact family behind each group of results. Repository continuous integration checks the theorem-count floor, rejects admitted proofs in the public library, and rebuilds the finite certificates used here; simulator replay is governed by the simulator receipts and reproduction commands. Exact file paths, manifests, receipts, mutation suites, and negative controls accompany each module in the public repositories [1, 2]. Online Resources 1 to 3 supply compact snapshots of the formal library, the exact certificates, and the finite evidence cited in this table.

Table 4: Principal verification artifacts and their scope.

Results	**Principal artifacts**
Normal forms, stability, refinement, complexity, and the transactional diamond	The observable-normal-forms library and component-paper reductions; the local-diamond and Newman arguments of this manuscript; the finite transaction-model verifier
Born, Lüders, Tsirelson and its saturation, no-cloning	The event-algebra modules for the interface, conditioning, the bound, partition and state maps, and the public record algebra, with per-module axiom audits
Rank-three completion, Coulomb–Green, seam holonomy, and the local face action	The port-Gram repair band, covariance and frame-quotient modules with a source-pinned receipt and independent verifier; the discrete Coulomb–Green and local-face Maxwell modules
Edge-center entropy and central defects	Overlap-cocycle and one-sided reduction proofs in this manuscript; the Einstein-branch entropy module; the identity-channel collar model with its no-go boundary
Gauge Lie type, matrix witness, matter-kernel image, and exterior selection	The screen modules for the carrier action, commutant, incidence response and holonomy bridge; the exact-arithmetic port current certificate; the lattice quotient, trace-balanced kernel and exterior scan modules

Table 4 continued

Results	Principal artifacts
Mass shell, internal-clock identity, inertia slope, and the derived action	The mass-shell, internal-clock, composite-momentum, proper-time action, port-charge coupling, modular-additivity and Gibbs-reading modules; the path-Gibbs, Euler–Lagrange, Noether and enrichment-selection modules; the calibration import with its non-forcing receipt
Coupled matter continuum, interacting Hilbert space and model duration	Uniform-refinement and energy-estimate proofs; kinetic-metric reduction, completeness and operator-domain proofs; charged self-reading execution and configuration-clock receipts with independent verifiers [10]
Null tomography, small-ball arithmetic, and tensor completion	The Einstein-branch tensor module for null directions, design, decoder, injectivity and metric ambiguity; the analytic constants of this manuscript; explicit null-balance, Ward, Bianchi and scale premises
Read-law necessity, crossing-area limit, native transport and vacuum fidelity	The M1 necessity, interface, quantum-transport and vacuum-fidelity reductions with their axiom audits and exact executable controls [11]
Angular-rank theorem, frozen branches, and the dispersion band	The primitive-port prediction and carrier-class band modules; the multipole, universality, prediction and band receipts with mutation tests; the custody-bound prediction register
Dark sector, the vacuum coordinate, and the closure roots	The dark-sector, deep-profile, per-cut, collar-premise and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root and the posterior propagation record
Repair gap and the conditional Yang–Mills identification	The Yang–Mills gap, witness and repair-gap-chain modules; the collar gap certificate and the finite transfer receipt with its test
Exact federation under the canonical law	The level-six configuration, seeds, seam arrays, loads, per-schedule V ledgers, quotient hashes, response kernels, standalone verifier and public archive; the level-five to level-ten settlement receipts with host verification records and settling readouts
Closure loop at carrier scale	The event logs of the canonical, integer, overwrite, tetrahedral and octahedral sources, the recovered specifications and invariant vectors, producer-free verifier, and public archive
Causal poset	The source-record family receipts with exact all-pairs counts through $q = 89$ and the sampled $q = 144$ level, the carrier realization receipts and event logs, the chain-count and interval-spectrum receipts, the standard-library generator that rebuilds the poset from its definitions, checker, and public archive

13 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes, and decision rules before eligible comparison [3]. Propositions 10.4 and 10.5 are the prospective discriminators in this paper; their coefficient rays are fixed separately, their excluded exposure classes are declared, and a qualifying failure rejects the tested physical propagation branch, with OPH-wide scope requiring a stronger source law that supplies the physical action, readout, frame transport, and exclusivity theorem for that branch. The Einstein composition requires one physical refinement tower satisfying its event, modular, stress, entropy, continuum, vacuum, and scale premises together, and the gauge and matter conclusions require laboratory-current attachment and a continuum operator limit; failure to construct those maps leaves the finite theorems intact

and removes their physical application. The remaining surfaces carry their own decision shapes. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; exclusion of $(w_0, w_a) = (-1, 0)$ in any direction fails the fixed-capacity branch while agreement with that point supports nothing, because it is also the ΛCDM null; and the Yang–Mills identification fails if the support-visible continuum construction, reflection positivity, or the uniform repair gap fails.

14 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [20, 21, 22]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an explicit input and contributes the completed-record surface on which the event, conditioning, expectation and correlation identities operate, together with the consensus theorem that defines completion. Four neighbors calibrate the record story. Quantum Darwinism locates classical objectivity in redundant environmental records [54], and the completed-consensus record plays the same public role with redundancy replaced by protected overlap agreement and an explicit repair dynamics; the cellular-automaton program shares the finite substrate [55], while OPH locates the quantum identities on the completed record surface rather than in an underlying ontological basis; Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [56], whose premises concern pre-consensus perspectives that the completed surface does not represent; and relational quantum mechanics treats physical facts as relational [52] while the thermal-time program reads time from modular structure [53], whereas OPH attaches public facts to protected overlap records and consumes normalized modular flow only on a typed common tower, with its local-diamond proof neighboring asynchronous agreement [47, 48].

The underlying convergence ideas also have close established precedents. Dijkstra's central-daemon self-stabilization separates recovery from continued legitimate operation [49]. Fixed linear sheaf diffusion on finite inner-product spaces projects onto compatible sections [50]; with pinned boundary data, harmonicity alone need not remove all disagreement. These are conditional construction tools. A candidate must supply their restriction maps, state spaces and admitted schedules; convergence or shared fixed states alone does not identify a microscopic physical law.

The gravitational branch belongs to the family of thermodynamic and entanglement-based routes to geometry [63, 64, 67], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [60, 61, 62]. The contribution of Theorem 7.2 is the explicit composition contract, with event, modular, stress, entropy, asymptotic, vacuum, and scale premises in one statement and the finite null ambiguity identifying exactly where a metric-proportional term enters; the route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [65, 66], and what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section contains two reconstruction routes and one conditional global-form witness: Doplicher–Roberts/Tannaka reconstruction supplies the structural sector group [71, 72], complete A1 response and endogenous A2 transport force the local Standard Model gauge Lie algebra, and the explicit matrix map with the exterior-module scan gives the conditional packet group, with equality of the sector and packet groups a separate commuting-square premise.

A useful comparison of unification proposals asks how much physical structure follows from

each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

15 Conclusion

The finite architecture relates records, spatial carriers and matter through explicit mathematical interfaces. Its normal-form theorems control agreement, stability and refinement; finite effect valuations yield the probability rule for quantum records; carrier completion gives a three-dimensional position readback; complete response with endogenous transport forces $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$; and on the declared matter menu exhaustive selection yields an anomaly-free fifteen-state generation with common $\mathbb{Z}_6$ kernel. A supplied charged-scalar/Maxwell action connects these objects to a controlled nonlinear continuum trajectory and an interacting quantum state space, with first-order convergence of its real quartic-wave sector under uniform refinement and a unique self-adjoint Hamiltonian on the fixed mesh under the declared quantization; bounded self-reading patches execute the charged model. A source-selected action and one common physical realization are additional requirements.

The gravitational branch assumes a common physical refinement tower with the stated stress, entropy, continuum and scale data; null tomography, Ward conservation and the Bianchi identity then yield the Einstein equation up to a cosmological constant, and constructing that tower with the gauge and matter readouts is a separate requirement. Modular charge carrying no electromagnetic readout classifies a non-luminous gravitating sector whose deep-regime law is unique under scale covariance and quadrature composition, carrying one constant into flat rotation curves and $v^4 = GM_b a_0$; reading the dark-energy density as $\rho_{\mathrm{DE}} = \kappa/N$ sends fixed capacity to $(w_0, w_a) = (-1, 0)$; and on the support-visible compact-gauge branch the continuum Yang–Mills Hamiltonian gap is the limiting repair gap under the named certificates. The closure formulas are retrospective diagnostics of declared maps. The prospective content is the frozen dispersion surface, where $C_4 < 0$ with $B_0/C_4^2 \geq 10/21$ and B_6/B_0 inside $[-16/135, 16/75]$ excludes the whole declared carrier class at once when a resolved measurement falls outside it.

The finite theorems establish implications on their stated domains; the quantitative formulas are diagnostics or prospectively fixed tests. The five maps of Section 1.3 are consumed rather than constructed, so these results do not establish that the architecture describes our universe. A physical reconstruction requires one realization preserving the proved relations across records, fields, states and scales. The closure hypothesis of Section 1.5 proposes that such a self-consistent structure can determine its own physical content; the conditional theorems specify what that proposal must realize.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2048) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [3]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints

on observable normal forms, finite event algebras, consensus, Einstein reconstruction, and gauge structure [5, 6, 7, 11, 13], a relationship disclosed here so that overlap is visible.

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Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Mario Poneder contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the mathematical constructions, proofs, and counterexamples for the mechanism-variant comparison kernel and conditional example, protected-behavior first-hit obstruction profiles and finite-state transport, source-derived phase-effect selection, canonical adaptive-repair endpoint stratification, cumulative attempt-capacity bounds and sharpness results, and input-independent fixed computation-federation progress and execution. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to Tower endpoints, and contributed their manuscript, claim-registry, ledger, and reproducible-build surfaces. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

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Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation

repositories [1, 2]. Individual receipts and their bundle documentation provide the source and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive. Online Resource 3 supplies that evidence as a hash-bound archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 12. Online Resources 1 and 2 supply the formal and exact-computation sources as compact snapshots, and Online Resource 4 supplies the component preprints.

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